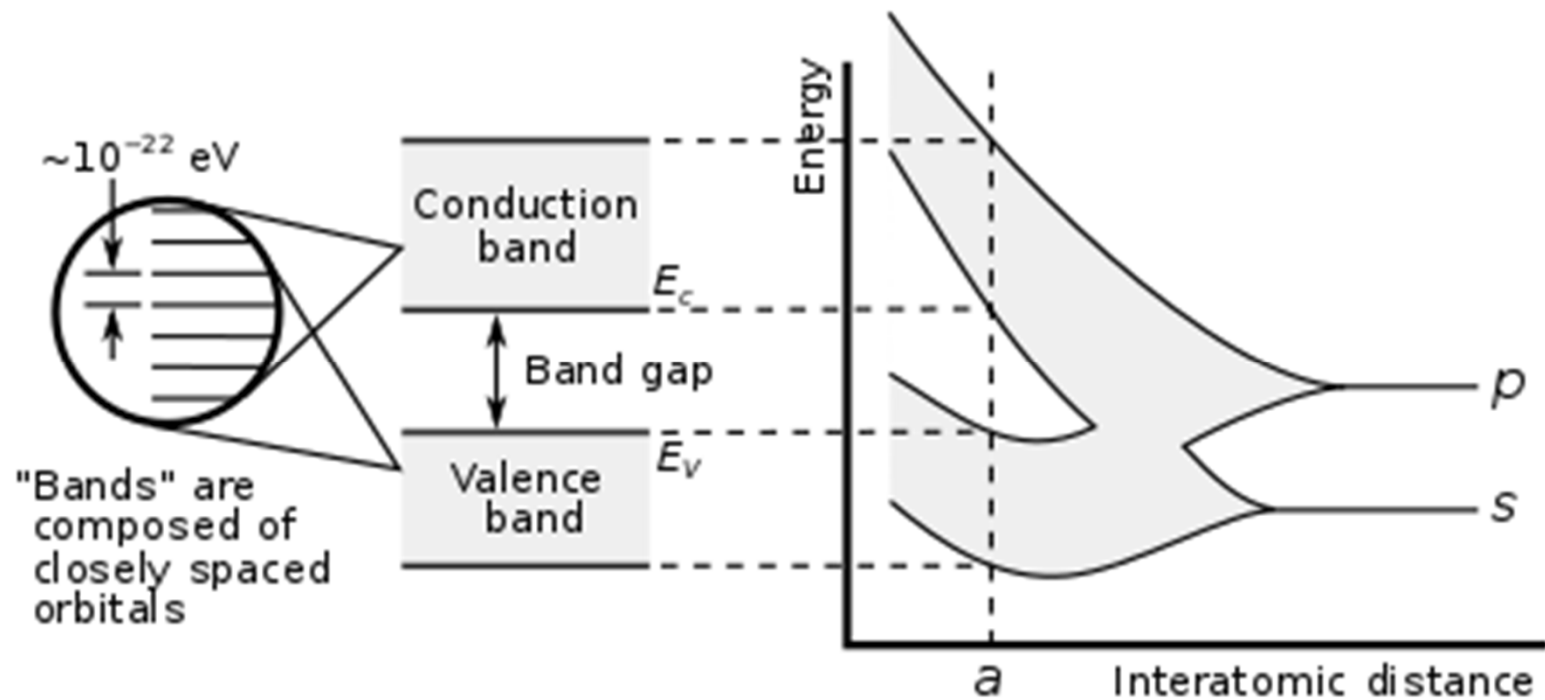


L3: The Energy Bands, Density of States



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Quantum Mechanics

Electron in a finite space $\rightarrow$ energy is quantized.

Pauli exclusion principle $\rightarrow$ one quantum state can be occupied by only one electron

Can we extend these concepts to electron in a crystal lattice?

Formation of Energy Bands

When two hydrogen atoms are brought close enough for the wave functions of $n=1$ electrons to start interacting, the $n=1$ state splits into two different energies, in accordance with Pauli exclusion principle.

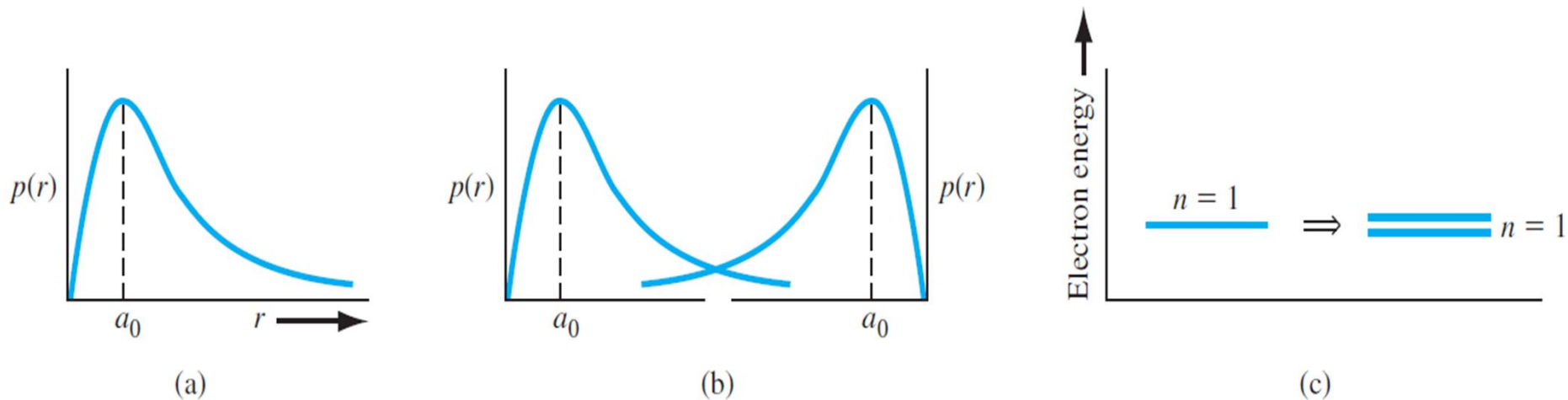
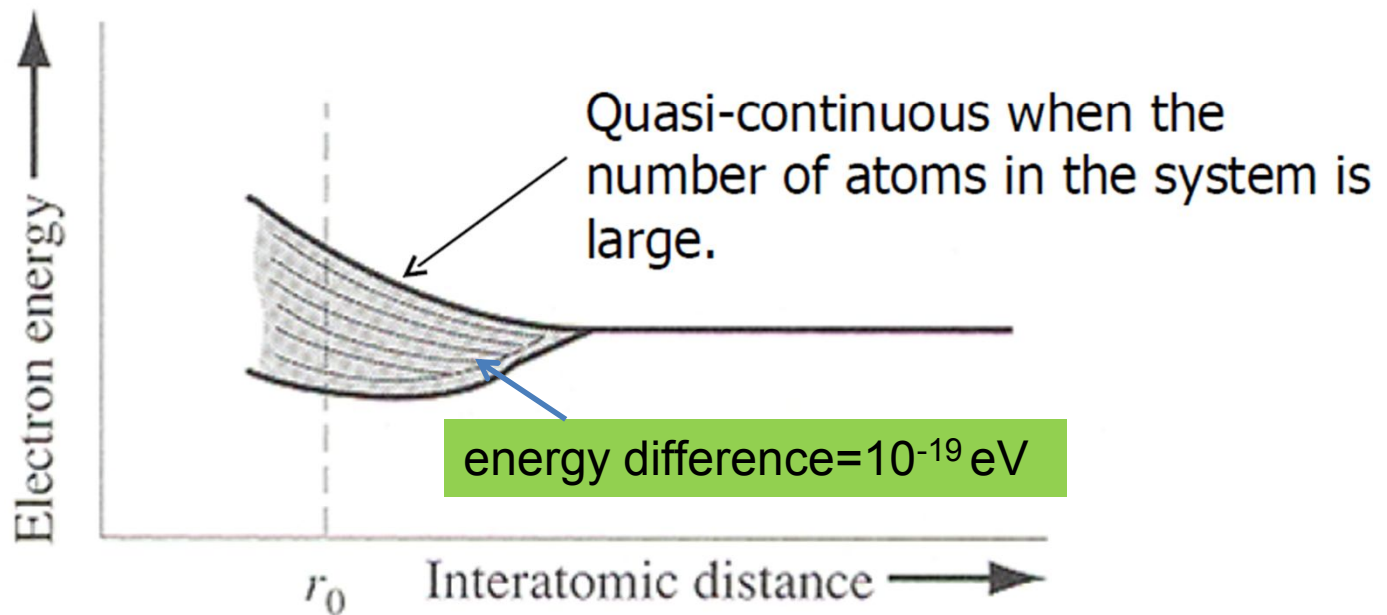


Figure 3.1 | (a) Probability density function of an isolated hydrogen atom. (b) Overlapping probability density functions of two adjacent hydrogen atoms. (c) The splitting of the $n = 1$ state.

Formation of Energy Bands

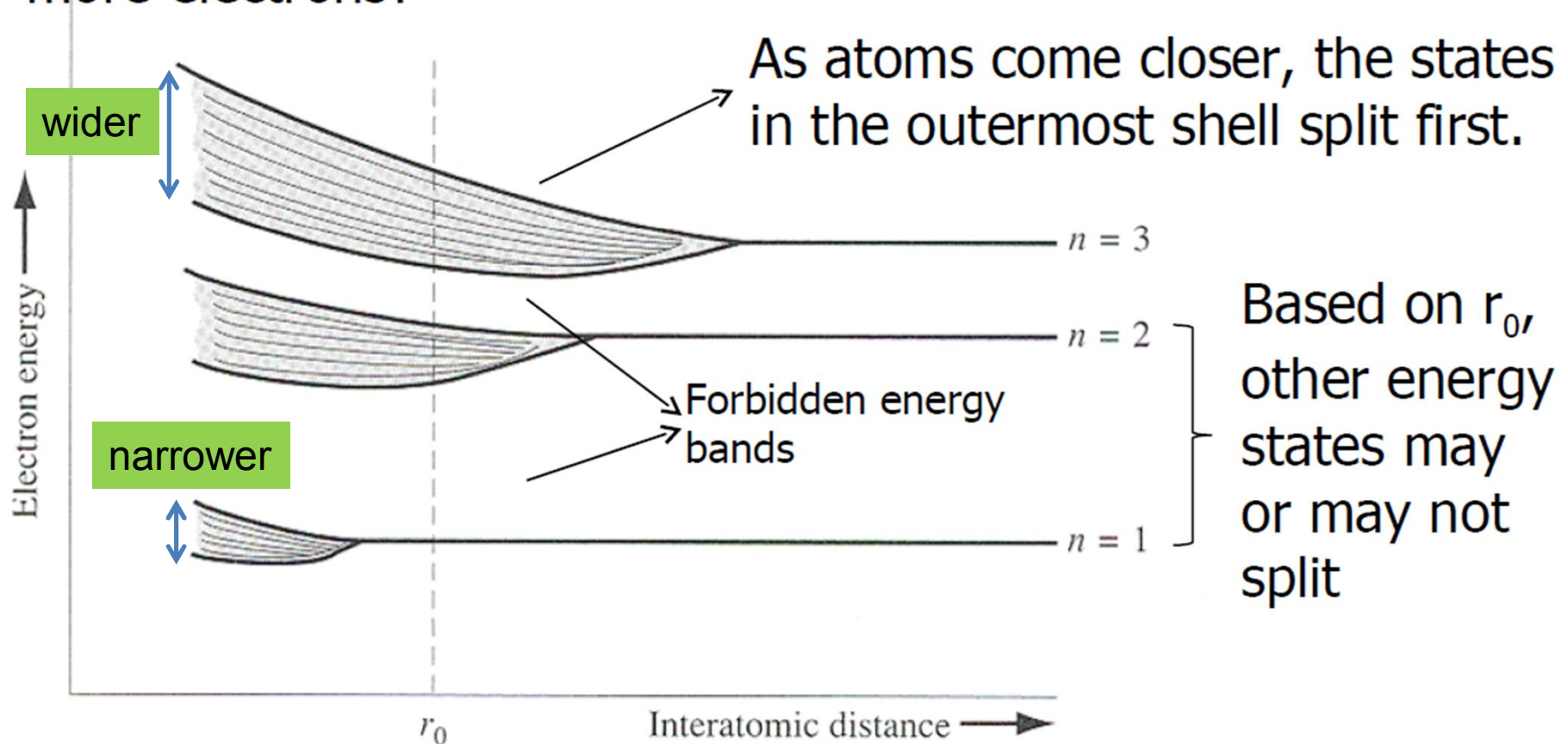
Hypothetically, if we have a periodic arrangement of many hydrogen atoms and they are brought close enough $\rightarrow$ initial quantized energy level will split into band of discrete levels.



The splitting of an energy state into a band of allowed energies (r_0 is the equilibrium inter-atomic distance in the crystal)

Formation of Energy Bands

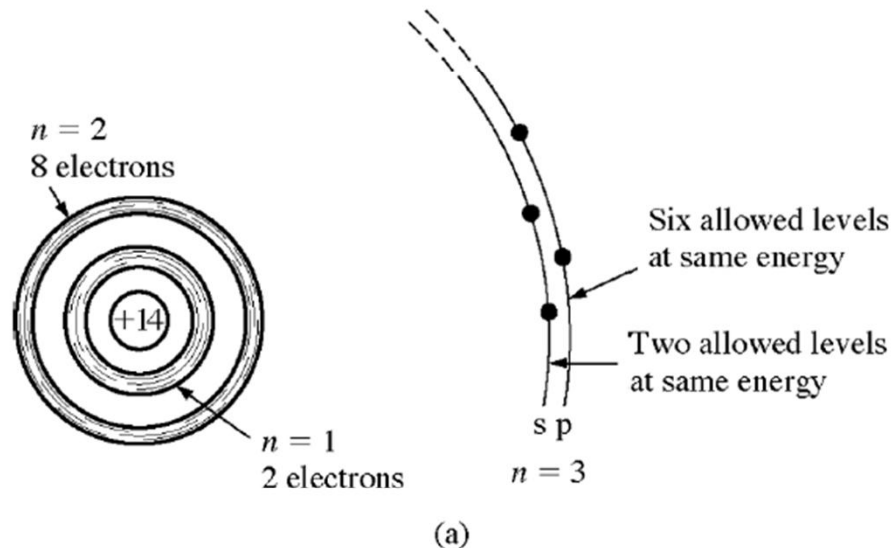
What happens in an atom containing many more electrons?



Splitting of energy states into allowed bands of energies in an atom containing electrons up to $n=3$.

Formation of Si Energy Bands

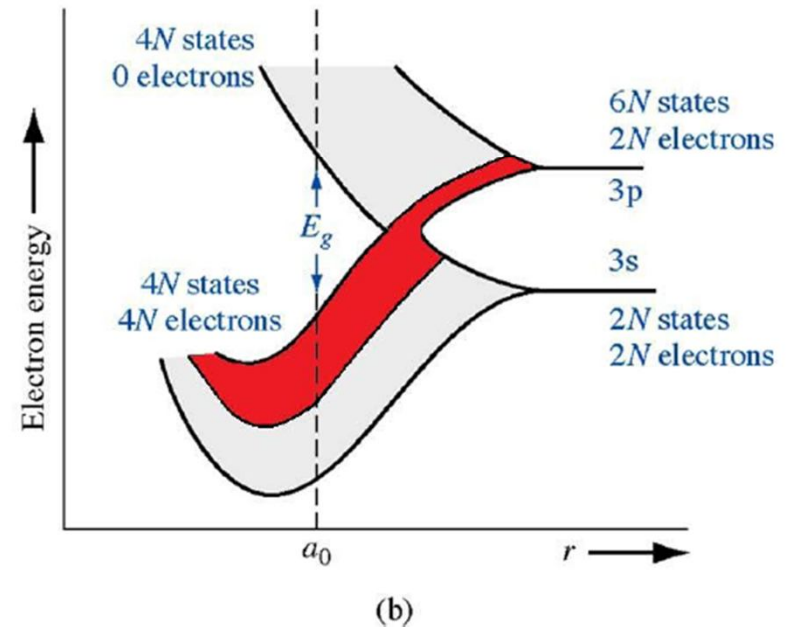
Actual splitting can be more complex as indicated for $n=3$ shell in silicon.



Isolated silicon atom.

For silicon, we need only consider the $n=3$ level because the first two energy shells are completely full and are tightly bound to the nucleus.

$T=0\text{K}$ lower (valence) band full, upper (conduction) band empty

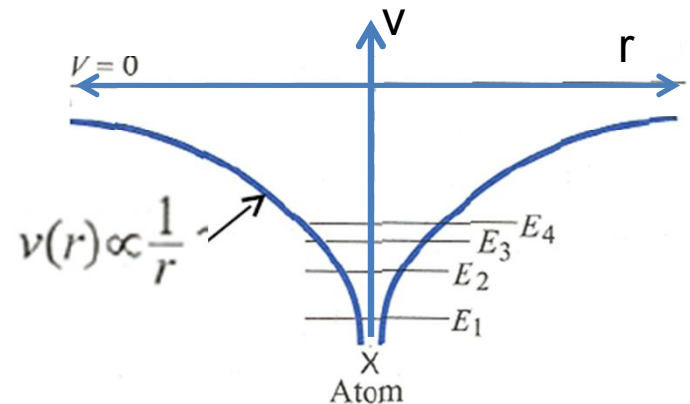


$2N$ 3s and $6N$ 3p states merge and form $8N$ new states, in which lower $4N$ states are occupied and upper $4N$ states are empty.

Kronig-Penney Model

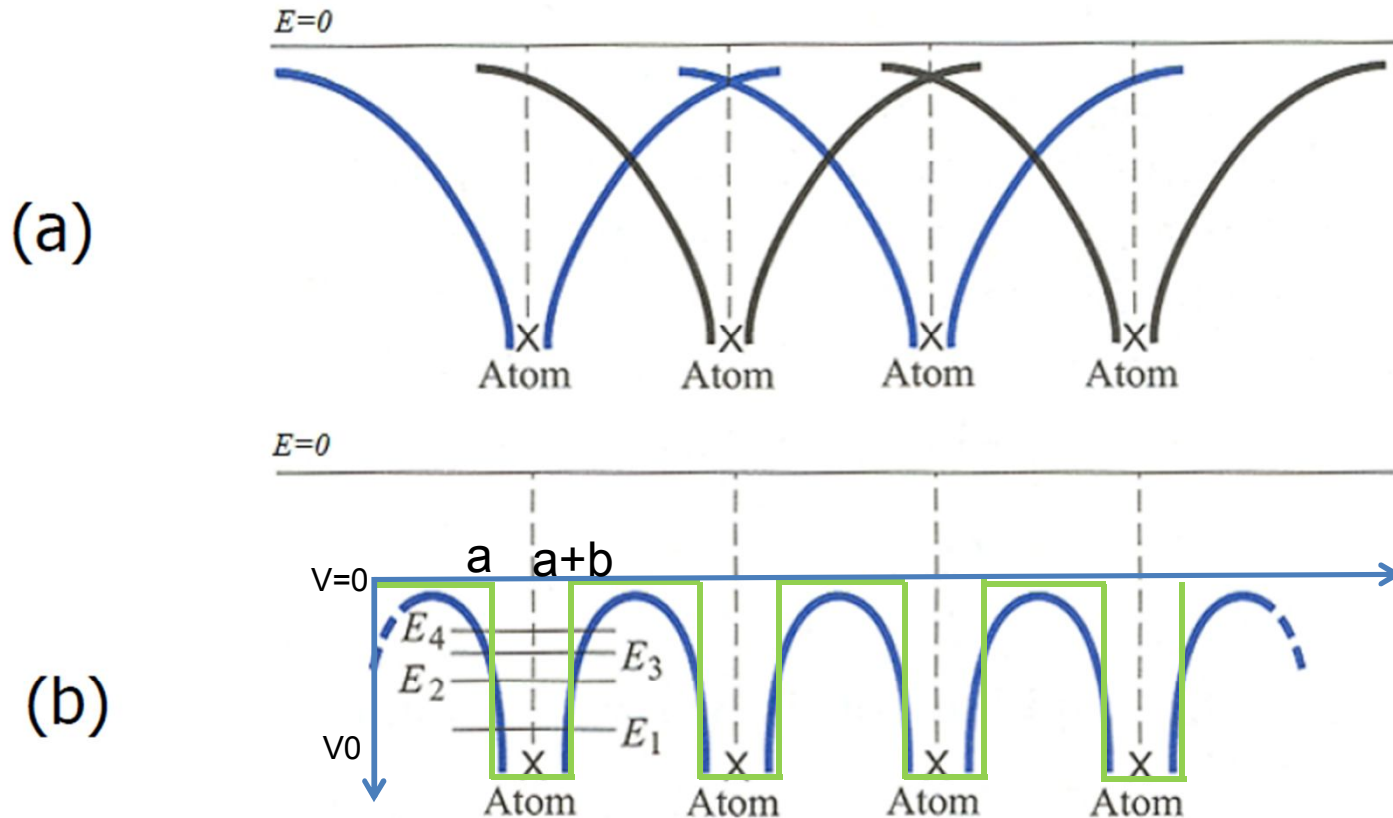
The band structure can be developed more rigorously by considering quantum mechanics.

- potential (V) is inversely proportional to distance from positively charged nucleus.
- electron is negatively charged.
 $E = -eV \rightarrow$ energy is negative (which means the electron is attracted to the nucleus)
- At infinite distance from nucleus both V and E are zero (free electron)



Potential and electron energy functions of a single, non-interacting, one-electron atom

Kronig-Penney Model

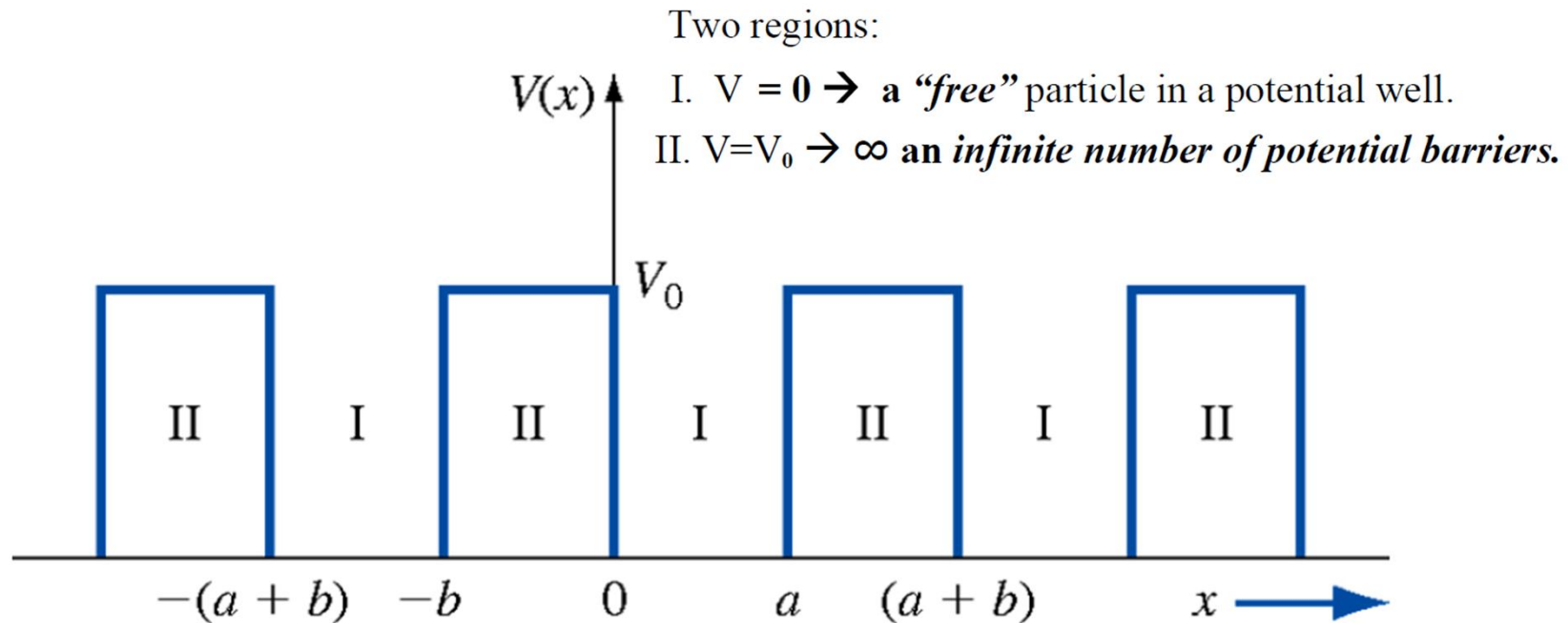


(a) Overlapping Energy functions of adjacent atoms (b) Net Energy function of a one-dimensional single crystal.

The potential functions of adjacent atoms overlap to form a one dimensional periodic potential function.

Kronig-Penney Model

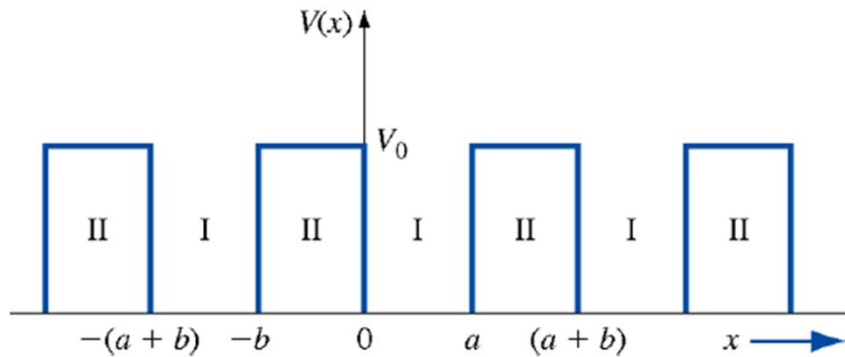
Simplified potential function $\rightarrow$ Kronig-Penney model



One-dimensional periodic potential function of the Kronig-Penney model

- **$a+b$** approximately equals to the distance between two adjacent atoms.
- Electron is bound in the crystal when $E < V_0$
- Electrons are contained in the potential wells.
- There is possibility of tunneling between potential wells.

Solving the Schrodinger equation



Time-independent Schrodinger wave equation

$$\frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2m}{\hbar^2} (E - V(x)) \psi(x) = 0$$

Region I : $V(x)=0$. Substituting $\psi(x) = u_1(x)e^{jkx}$

$$(1) \quad \frac{d^2 u_1(x)}{dx^2} + 2jk \frac{du_1(x)}{dx} - (k^2 - \alpha^2) u_1(x) = 0$$

$$\alpha^2 = \frac{2mE}{\hbar^2}$$

region II, $-b < x < 0$, in which $V(x) = V_0$

$$\psi(x) = u_2(x)e^{jkx}$$

$$(2) \quad \frac{d^2 u_2(x)}{dx^2} + 2jk \frac{du_2(x)}{dx} - \left(k^2 - \alpha^2 + \frac{2mV_0}{\hbar^2} \right) u_2(x) = 0$$

$$(3) \quad \text{The periodicity and the continuity condition: } u_1(a) = u_2(-b)$$

Solving the Schrodinger equation

$$P' \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$P' = \frac{mV_0ba}{\hbar^2}$$

- This is a relation between parameter k , total Energy E (through parameter α) and potential barrier bV_0 .
- This is not a solution to Schrodinger's equation but a condition for a solution to exist.

The above equation cannot be solved analytically. We have to use numerical or graphical methods to obtain **the relation between k , E , and V_0** . The solution will result in a band of allowed energies.

Solving the Schrodinger equation

$$P' \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

1st case, free electron:

$V_0=0$ hence:

$$\cos \alpha a = \cos ka$$

$$\longrightarrow \alpha = k$$

$$\alpha^2 = \frac{2mE}{\hbar^2} \quad \longrightarrow \quad E = \frac{p^2}{2m} = \frac{k^2 \hbar^2}{2m}$$

$$k = \frac{p}{\hbar}$$

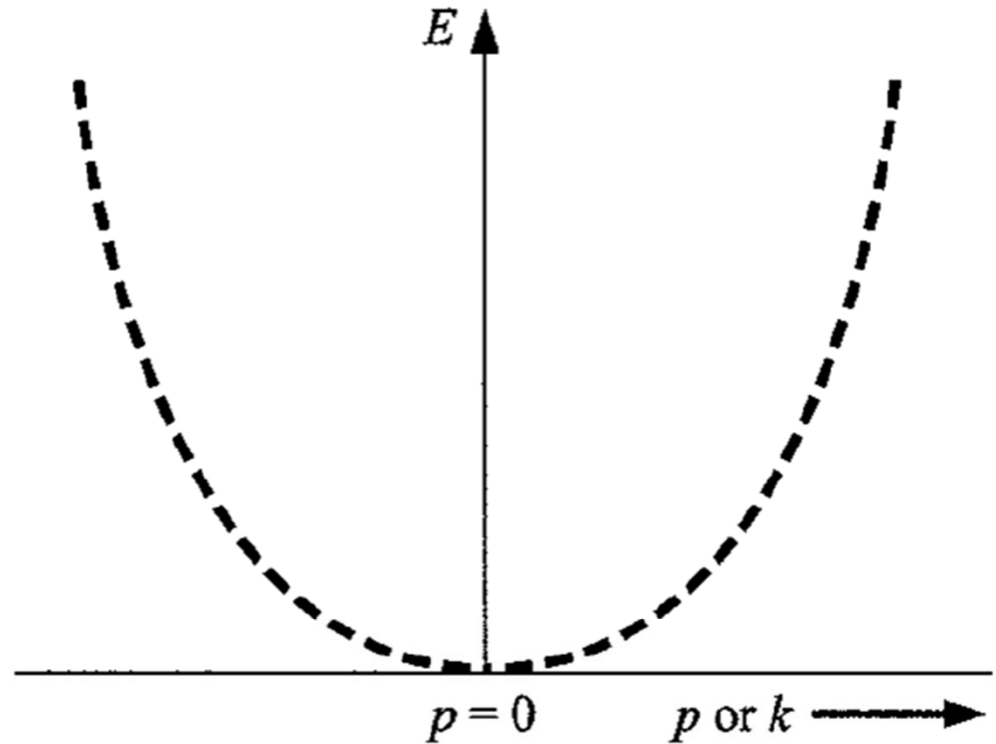
$\longrightarrow$ Constant of motion parameter k is related to the particle momentum for the free electron

(A result which we already know. This validates Kronig-Penny model)

E-K curve for free electrons

$$E = \frac{p^2}{2m} = \frac{k^2 \hbar^2}{2m}$$

$$E = KE + PE = KE$$



Parabolic E versus k curve for free electron

Solving the Schrodinger equation

Coming back to
Crystal-lattice $\rightarrow$

$$P' \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$\text{Let } f(\alpha a) = P' \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$$

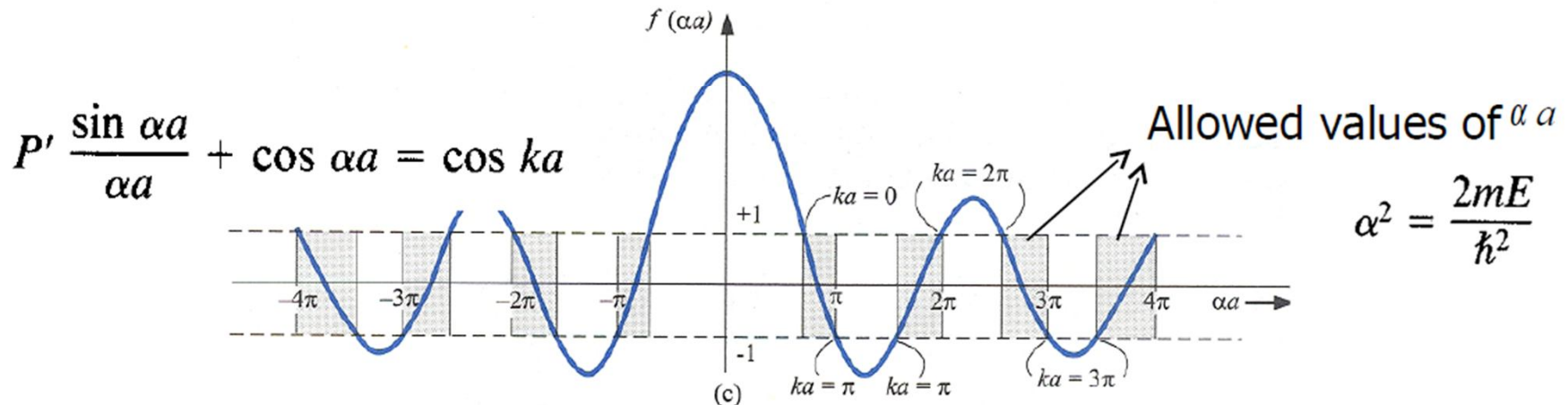
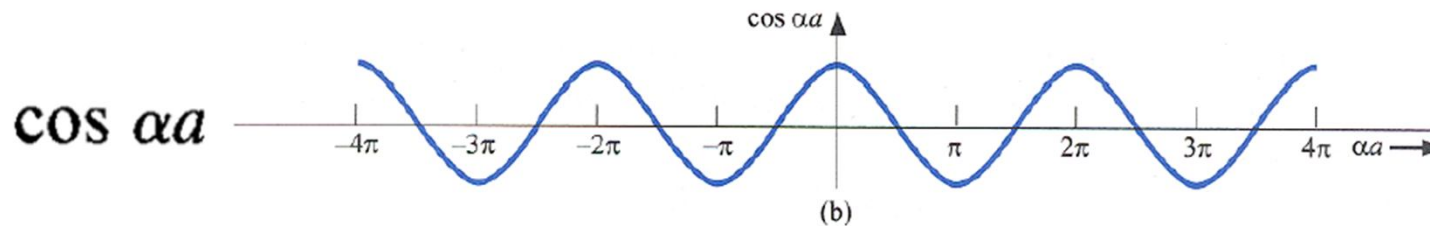
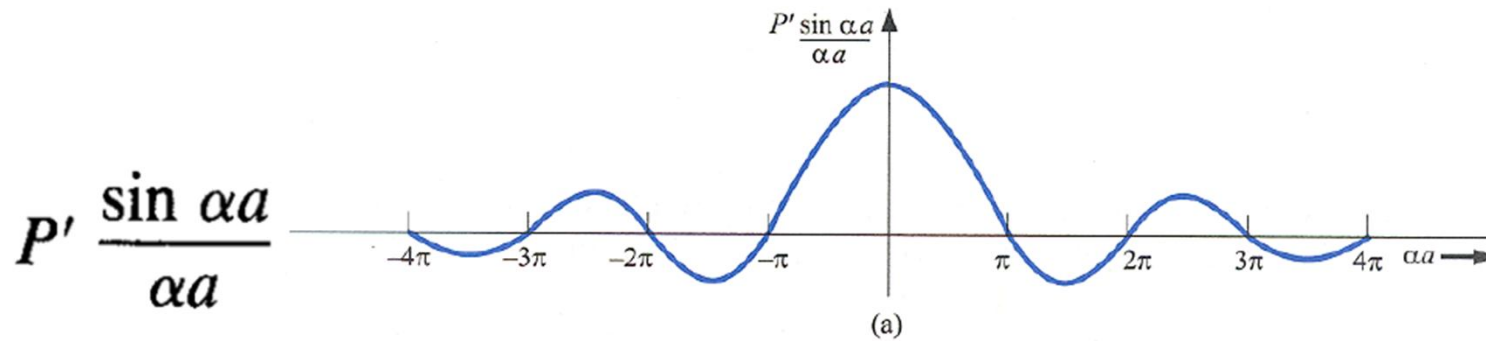
This function can only have values between -1 and +1, because it is equal to $\cos(ka)$

Hence αa can only have certain allowed values

Hence also the energy can have certain allowed values.

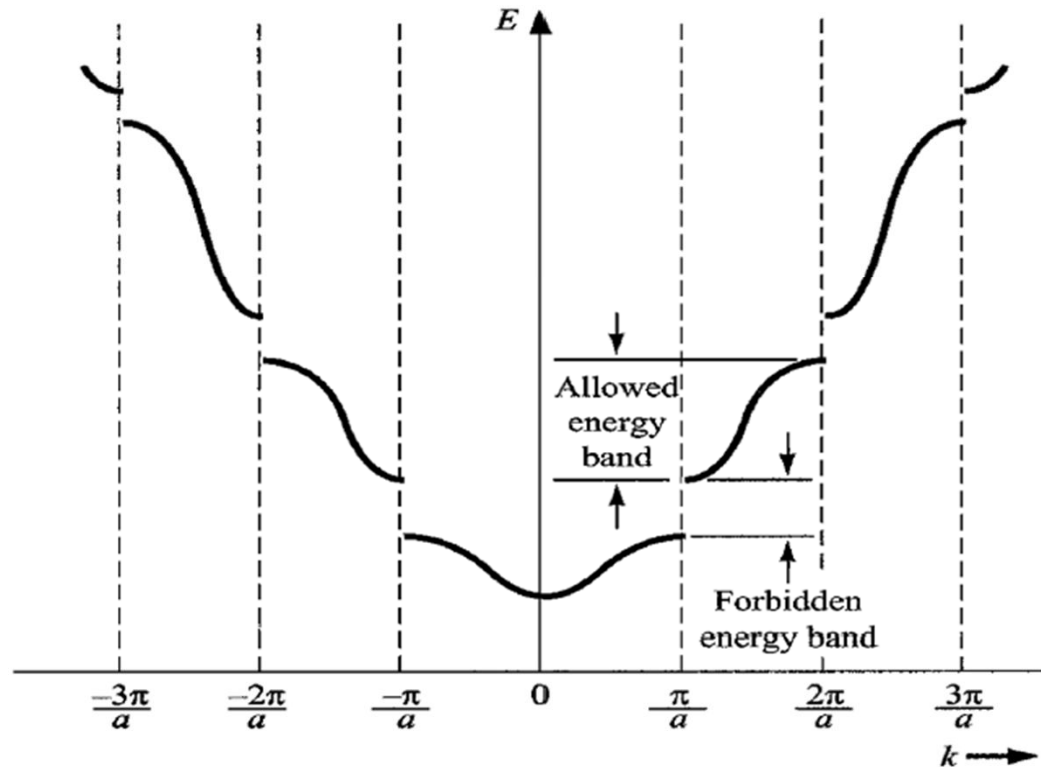
$$\alpha^2 = \frac{2mE}{\hbar^2}$$

Solving the Schrodinger equation



E-k diagram

Since E is related to k , we can make a plot of E vs. k .

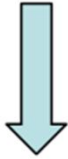


E versus k diagram generated from the figure in the previous slide.

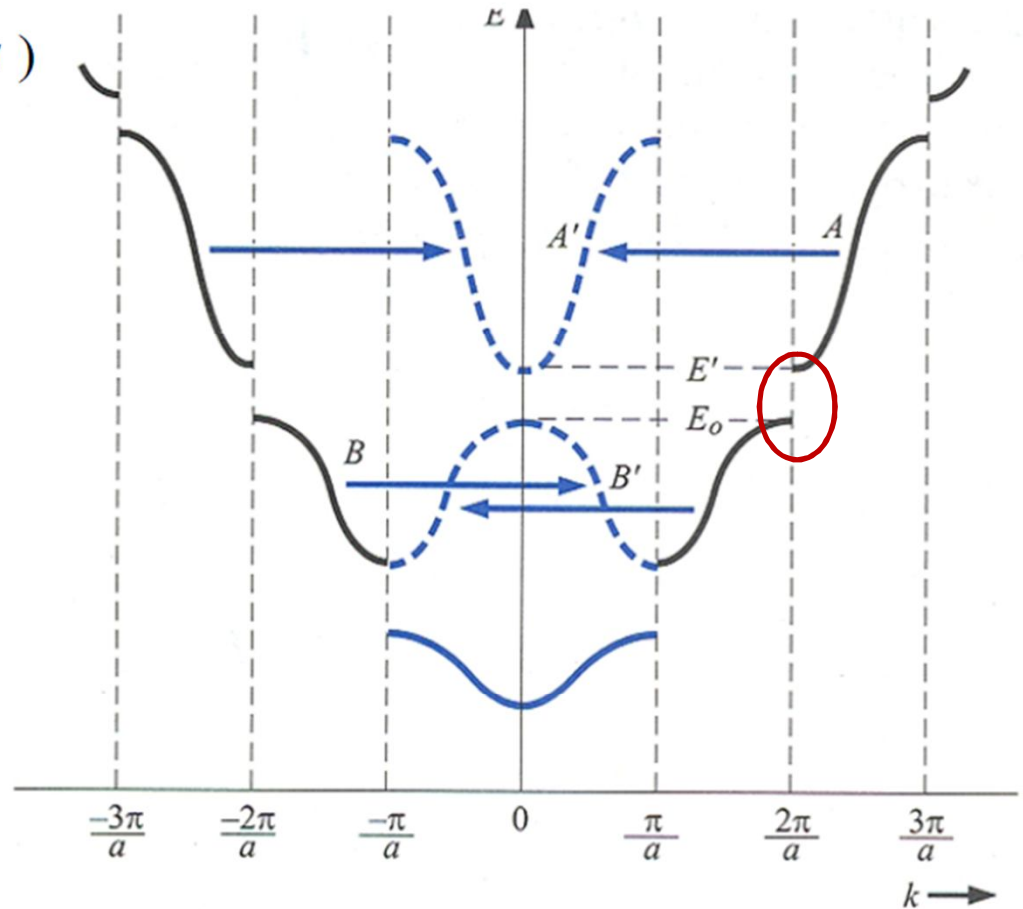
Note that E is discontinuous $\rightarrow$ there are allowed and forbidden energy bands.

Reduced E-k diagram

$$\cos ka = \cos(ka + 2\pi) = \cos(ka - 2\pi)$$



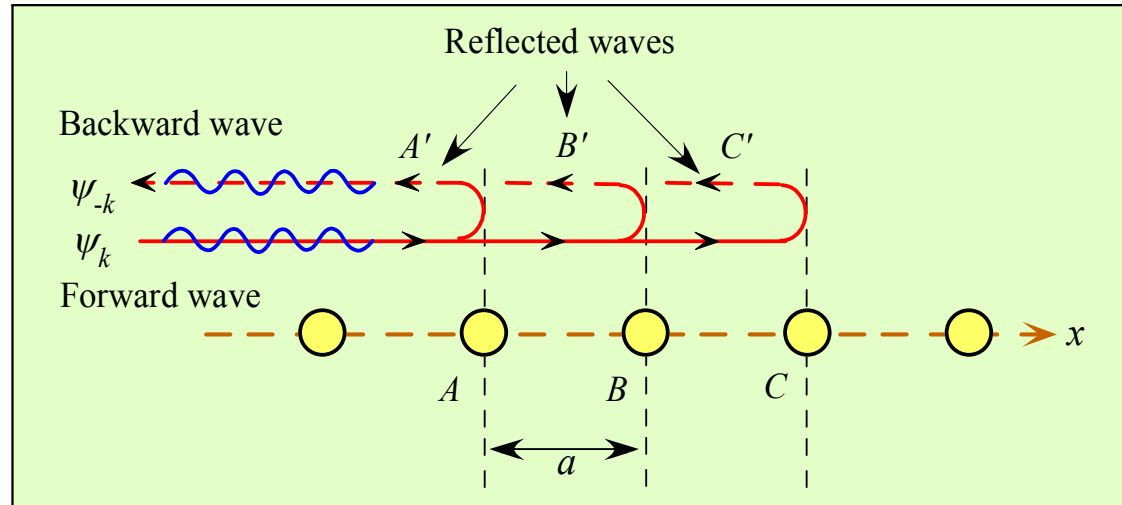
Various sections of the E versus k diagram can be shifted by $n \frac{2\pi}{a}$



E versus k diagram showing displacements by $n \frac{2\pi}{a}$ of several sections of the allowed energy bands.

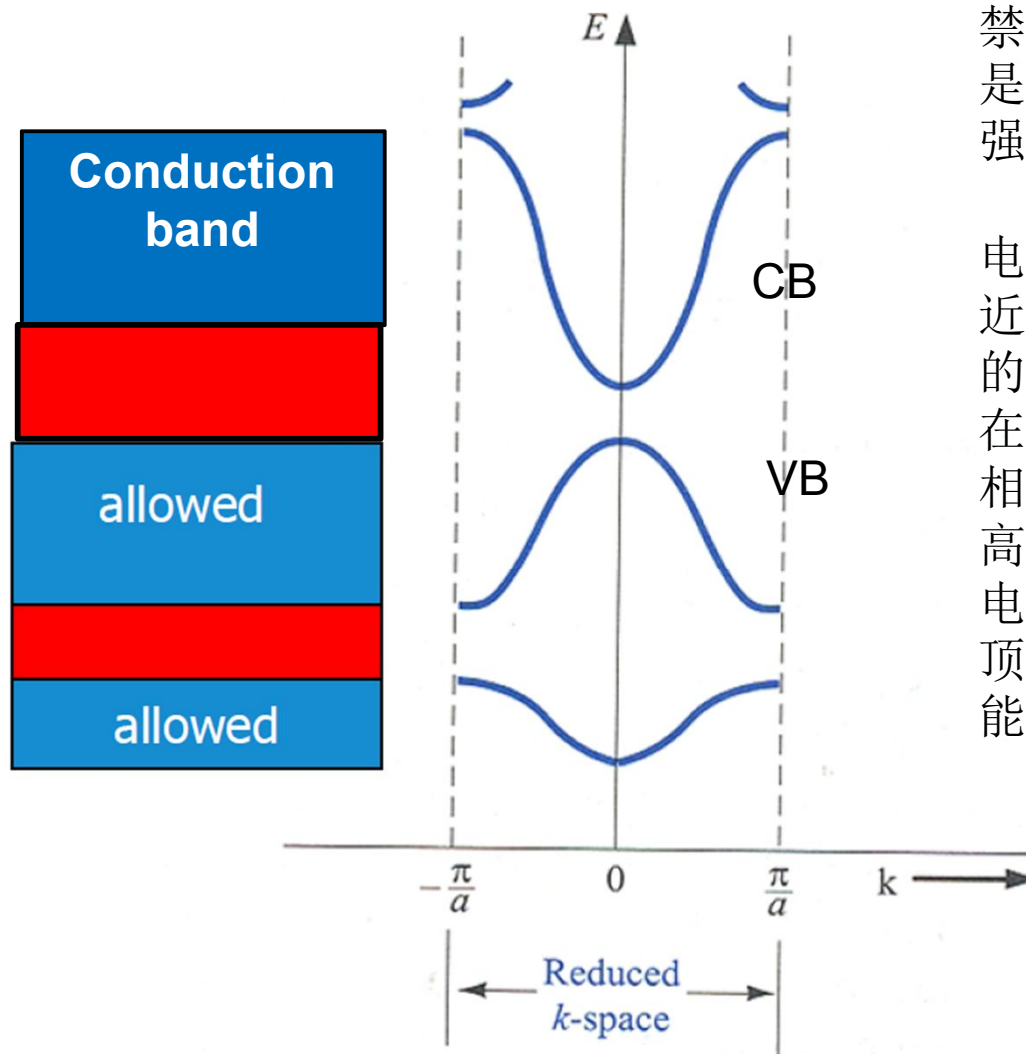
Reduced E-k diagram

- ❑ For electron move in a crystal, there is reflective wave and forward wave.
- ❑ Reflective waves interfere with each other constructive if:
$$2a = n\lambda = n\frac{2\pi}{K} \Rightarrow k = \frac{n\pi}{a}$$
- ❑ Combination of forward wave and reflective wave results in standing wave which does not propagate



An electron wave propagation through a linear lattice. For certain k values the reflected waves at successive atomic planes reinforce each other to give rise to a reflected wave travelling in the backward direction. The electron then cannot propagate through the crystal.

Reduced E-k diagram and energy band



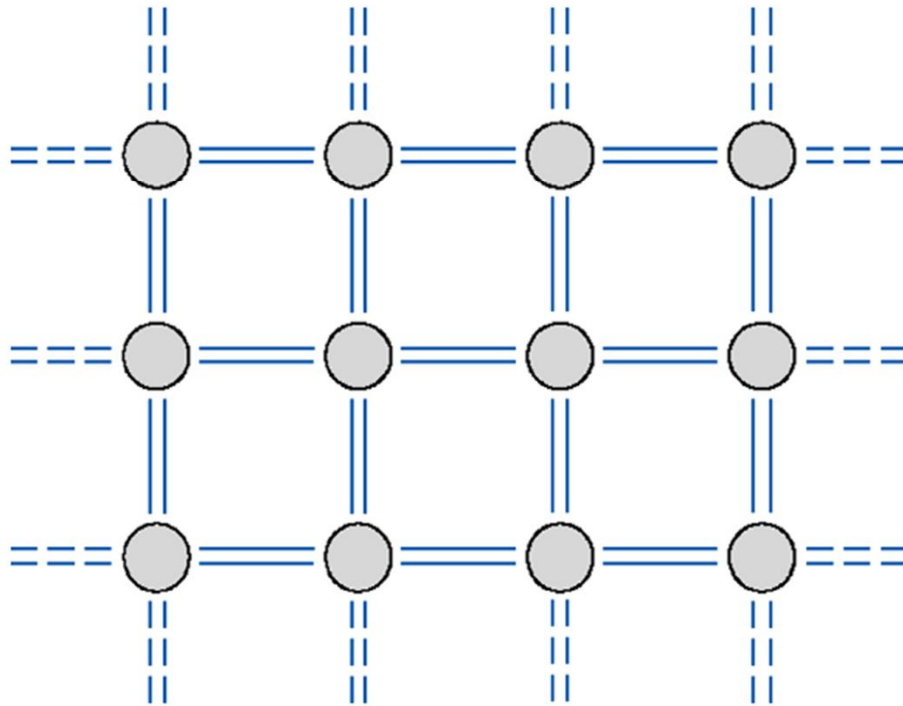
禁带宽度的大小实际上是反映了价电子被束缚强弱程度的一个物理量。

电子多分布在**导带底**附近（导带底相当于电子的势能），空穴多分布在**价带顶**附近（价带顶相当于空穴的势能）。高于导带底的能量就是电子的动能，低于价带顶的能量就是空穴的动能。

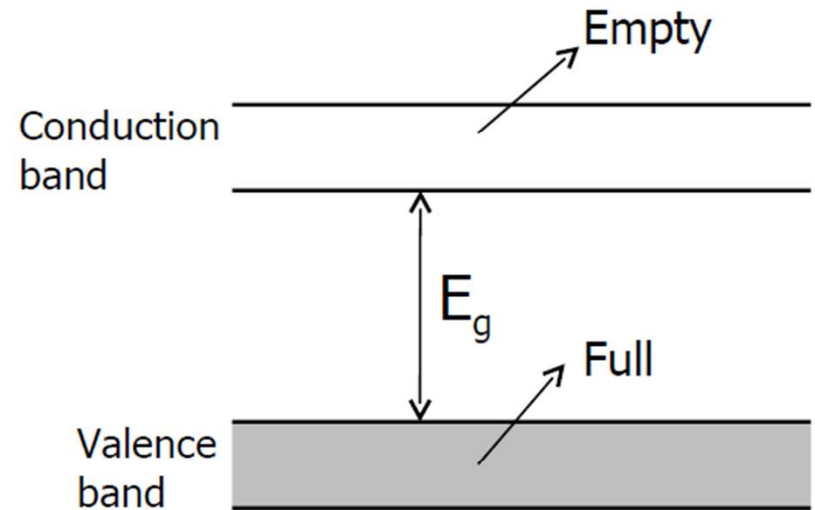
E versus k diagram in the reduced k-space representation

The Energy Band, the Bond Model and the Electrical Conduction

$T=0\text{K}$



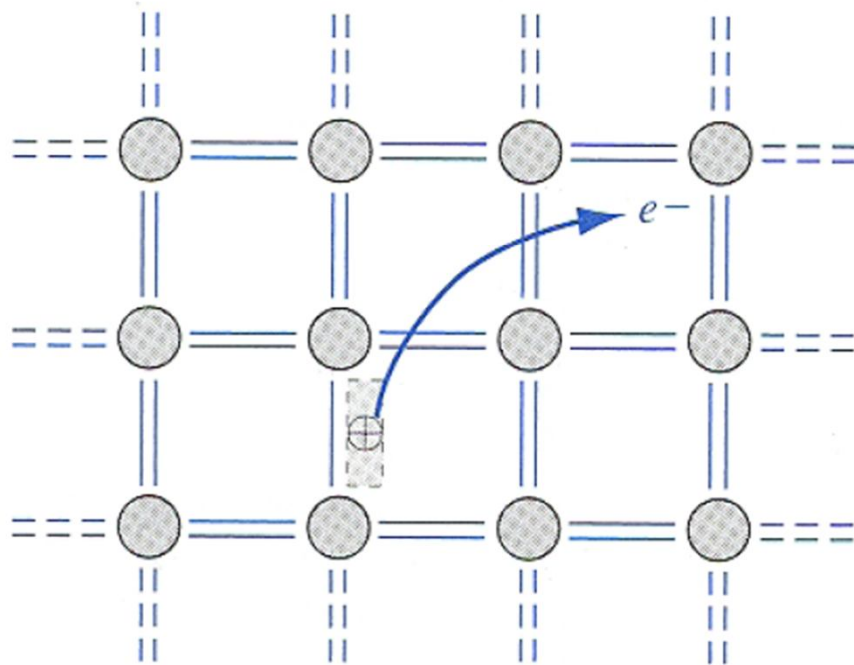
(a) Covalent bonding in Si



(b) line representation of energy band diagram

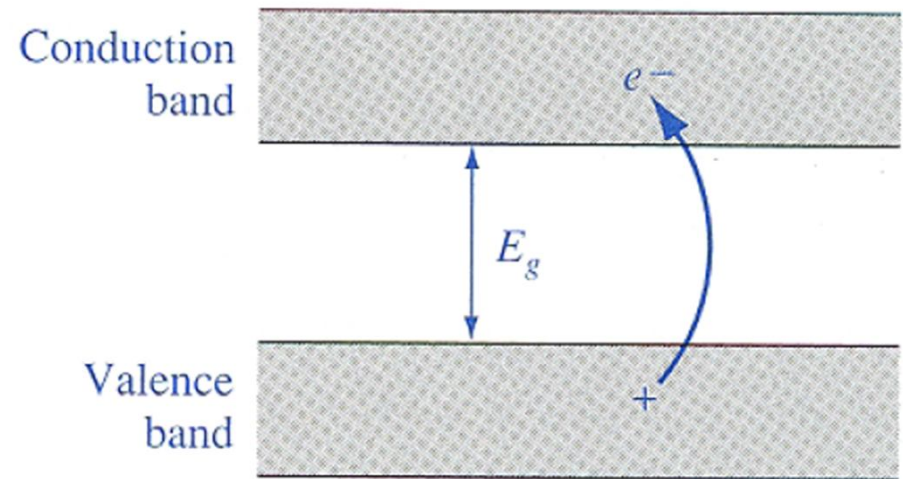
The Energy Band, the Bond Model and the Electrical Conduction

$T > 0K$



(a)

Two dimensional representation
of breaking of a covalent
bond.

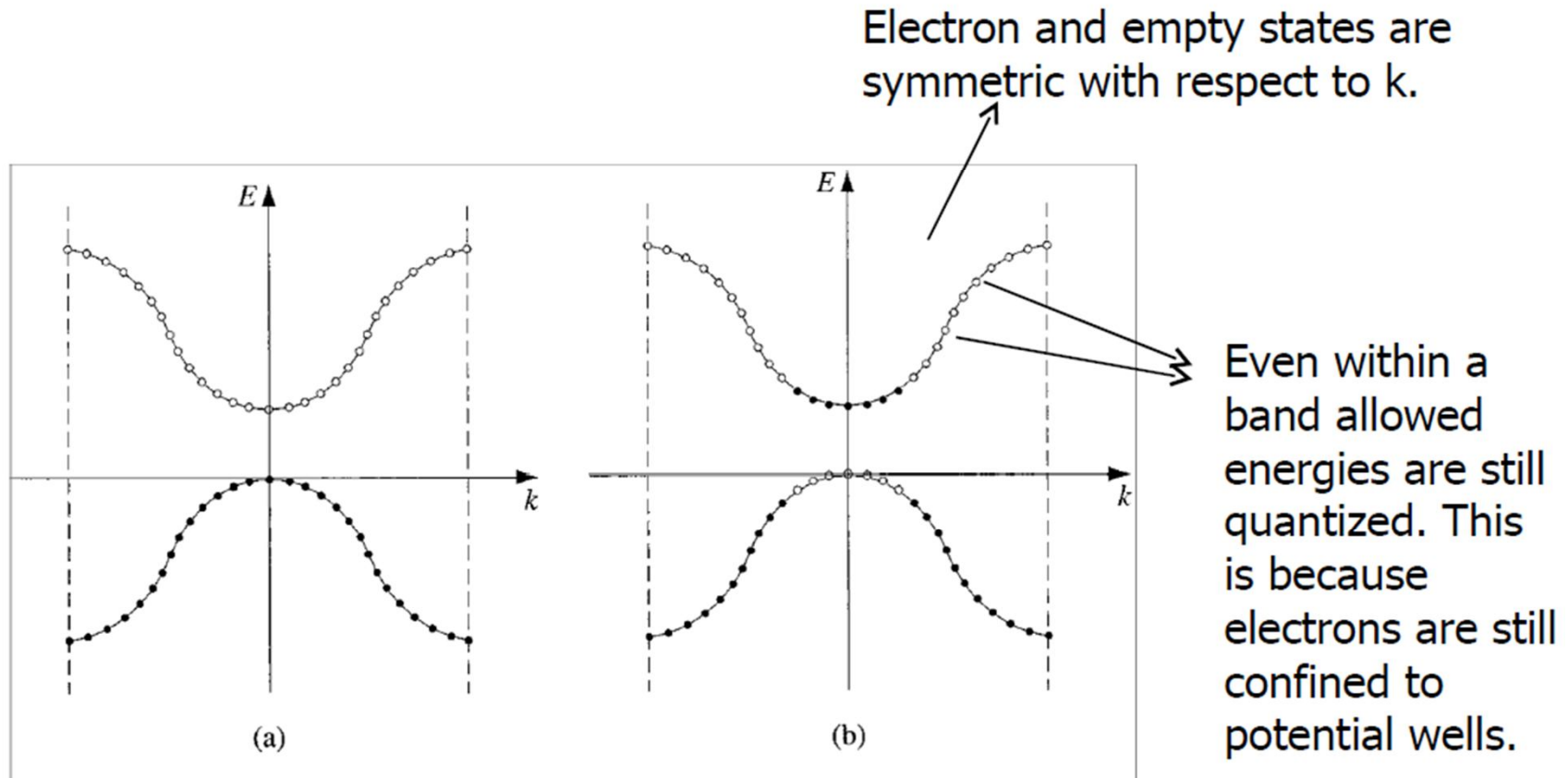


(b)

At $T > 0K$, some covalent bonds
break giving rise to positively
charged empty states and
electrons

The Energy Band, the Bond Model and the Electrical Conduction

E vs. k diagrams at $T=0\text{K}$ and $T>0\text{K}$ in the absence of external electric field.



The E versus k diagram of the conduction and valence bands of a semiconductor at (a) $T = 0^\circ\text{K}$ and (b) $T > 0^\circ\text{K}$

Drift Current

$$J = qNv_d (\text{A/cm}^2)$$

$q \rightarrow$ electron charge

$N \rightarrow$ charge density (C/cm^3)

$V_d \rightarrow$ drift velocity of electron

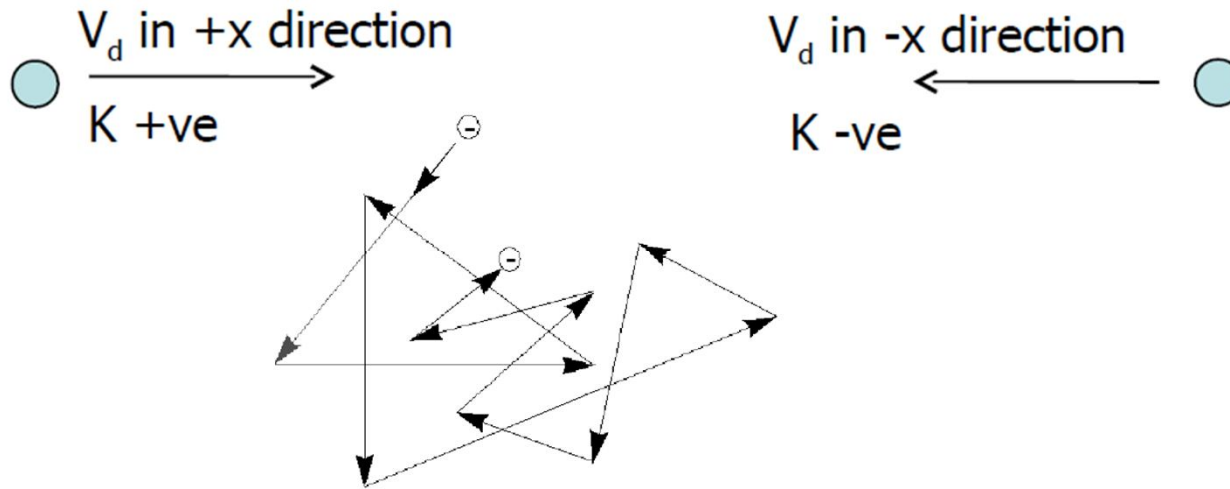
Considering individual
electron velocities



$$J = q \sum_{i=1}^N v_i$$

(summation taken
over unit volume)

Drift Current at zero field



No external
force



No of electrons with
 $+|k|$ value = No of
electrons with $-|k|$
value.



k related to
momentum.
So drift
current $J = 0$

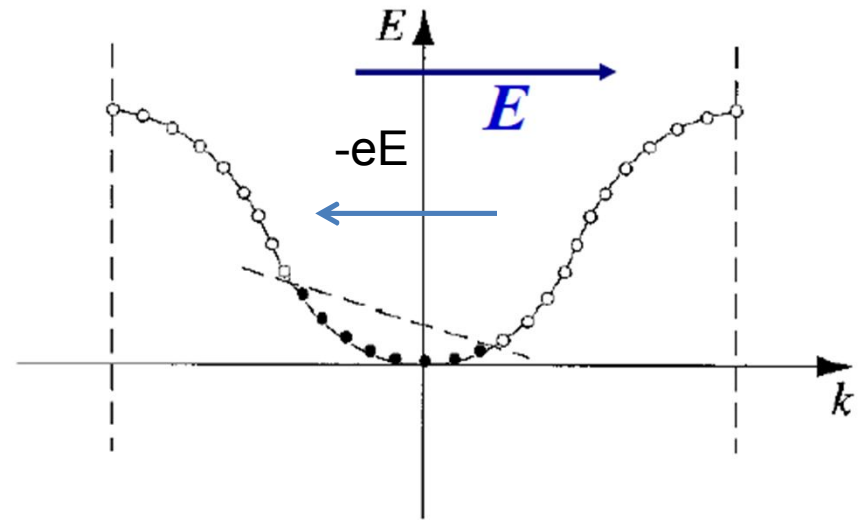
Drift Current

When ext force (electric field) applied → Electrons move into empty energy states in conduction band, gain net energy and a net momentum.

Drift current density due to motion of electrons

$$J = -e \sum_{i=1}^n v_i$$

(summation taken over unit volume)



Asymmetric distribution of electrons in E vs. k diagram when ext force is applied.

Effective Mass

Total force acting on electrons in crystal

$$F_{\text{Total}} = F_{\text{ext}} + F_{\text{int}} = ma$$

F_{int} → internal forces in the crystal due to +vely charged ions and –vely charged electrons

F_{ext} → externally applied electric field

m → rest mass of electron

a → acceleration.

Effective Mass

It is difficult to account for all of the internal force. So we define

$$F_{\text{ext}} = m^*a$$

m^* → effective mass of the electron (which takes into account internal forces)

Acceleration a → directly related only to ext applied force.

Velocity and E-K diagram

Free electron:

Taking derivative of E w.r.t k,

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}, \text{ since } p = \frac{h}{\lambda} = \hbar k \quad \longrightarrow$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m} = \frac{\hbar p}{m}$$

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{p}{m} = v$$

Velocity is proportional to the slope of the E-k curve

This implies that from E vs. k diagram we can get the velocity of the electron in real space.

Effective Mass and E-K diagram

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{p}{m} = v$$

Again taking derivative of E w.r.t k,

$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m}$$



$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m}$$

This implies that from E vs. k diagram we can also get the effective mass of the electron in real space. For a free electron, the second derivative and hence mass is constant.

Mass is reverse proportional to the curvature of the E-k curve

Effective Mass and E-K diagram

$$F = ma = -eE$$



$$a = \frac{-eE}{m}$$

Motion of electron is in opposite direction of the applied electric field due to the negative charge.

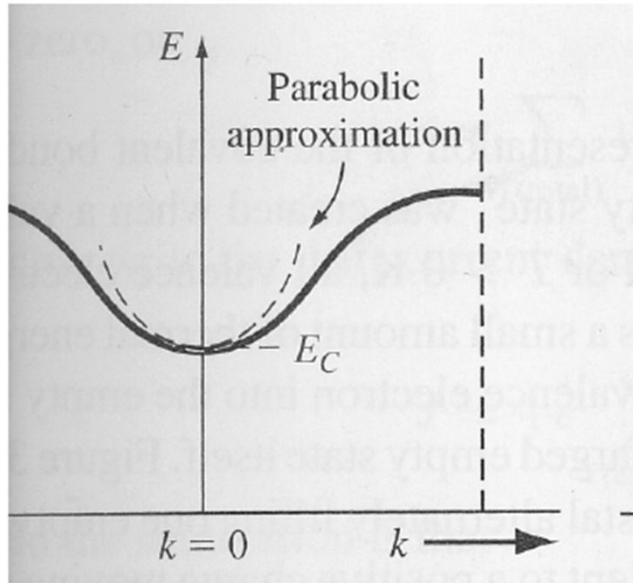
We now apply these concepts to the electrons in the crystal

Electron in conduction band

The energy of electrons at the bottom of the conduction band can be approximated as

$$E - E_c = C_1(k)^2$$

$$C_1 \rightarrow +ve$$



Conduction and valence bands in reduced k space with their parabolic approximations to compare with free electron

Electron in conduction band

$$\frac{d^2 E}{dk^2} = 2C_1$$



$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{2C_1}{\hbar^2}$$



$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{2C_1}{\hbar^2} = \frac{1}{m^*}$$

$$a = \frac{-eE}{m_n^*}$$

Effective mass:

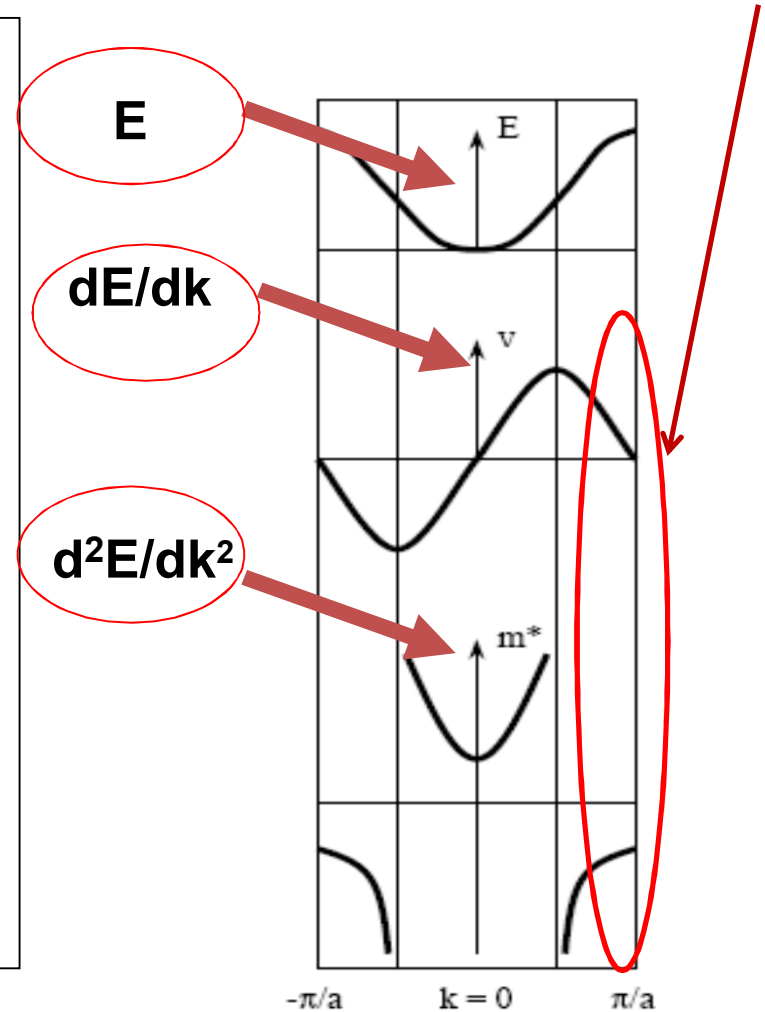
- connects quantum mechanics and classical mechanics
- effective mass varies with k , but almost constant at the bottom of the conduction band.
- positive value since C_1 is +ve

E-K diagram : Velocity and effective mass

At $k=0$, electron has a constant positive value and it rises rapidly as k value increases. After experiencing a singularity (infinite mass) the effective mass becomes negative up to the top

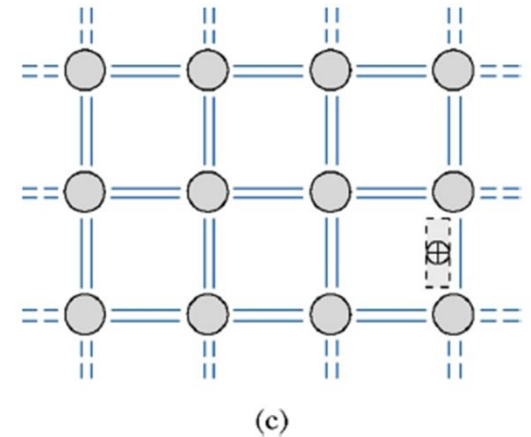
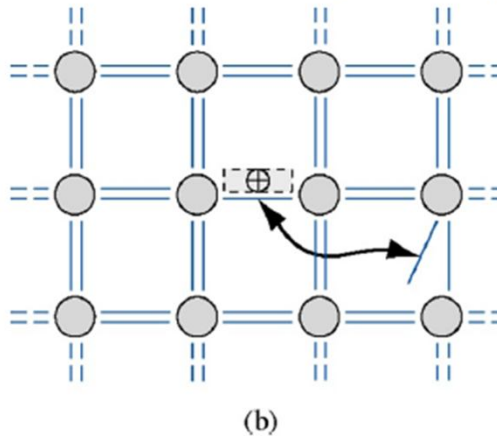
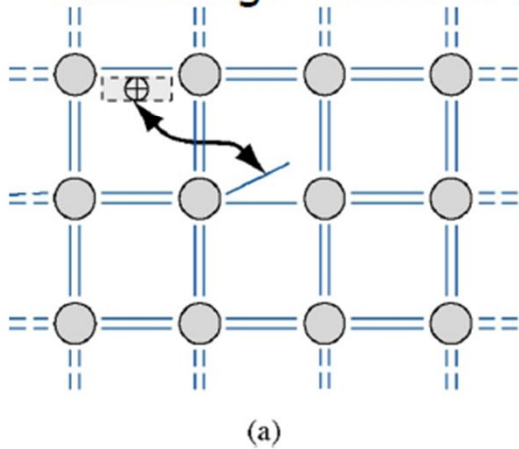
- Therefore:
- m^* is **positive** near the bottoms of all bands,
- m^* is **negative** near the tops of all bands.

Are the drawings correct?



Concept of the hole

- When valence electron goes to conduction band positively charged empty state is created.
- If a valence electron gets a small amount of energy, it can occupy this empty state.
- Movement of valence electron $\longleftrightarrow$ movement of +vely charged empty state in opposite direction.
- The charge carrier in the form of +vely charged empty state is called the hole.



Drift current due to electrons in valence band

Collective movement of filled electrons

$$\sum_{i(\text{filled})} v_i = \sum_{i(\text{total})} v_i - \sum_{i(\text{empty})} v_i$$

$$J = -e \sum_{i(\text{filled})} v_i \quad \longrightarrow \quad J = -e \sum_{i(\text{total})} v_i + e \sum_{i(\text{empty})} v_i$$

Also in a band that is completely full, electron distribution is symmetric with respect to k . The net drift current density generated from a completely full band is then 0.

$$\longrightarrow \quad -e \sum_{i(\text{total})} v_i \equiv 0$$

Drift current due to electrons in valence band

$$J = -e \sum_{i(\text{total})} v_i + e \sum_{i(\text{empty})} v_i$$

and

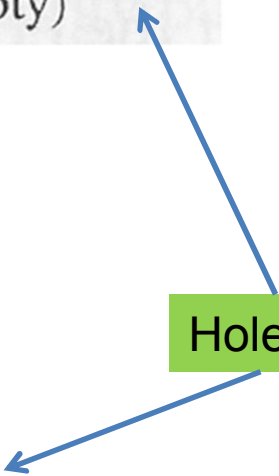
$$-e \sum_{i(\text{total})} v_i \equiv 0$$

Therefore

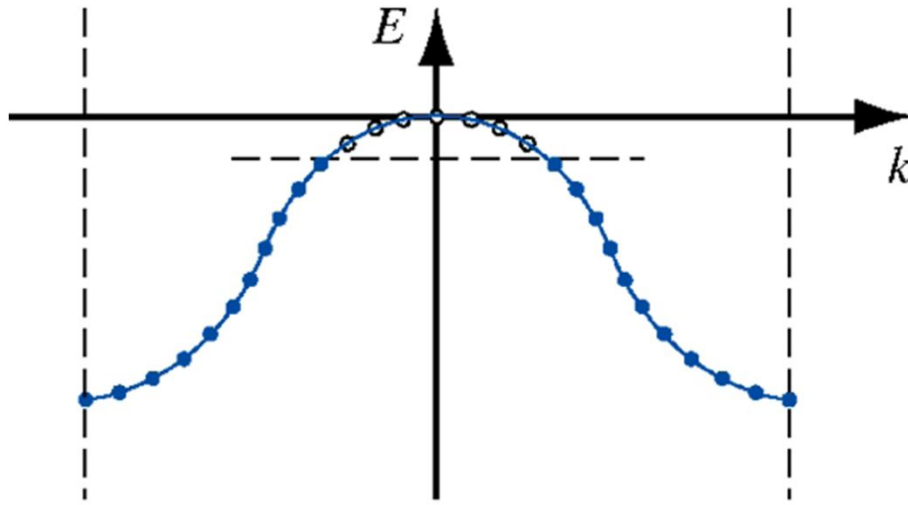
$$J = +e \sum_{i(\text{empty})} v_i$$

Drift current due to electrons in the filled state can also be looked at as that due to placing positively charged particles in the empty states.

Hole movement in valence band

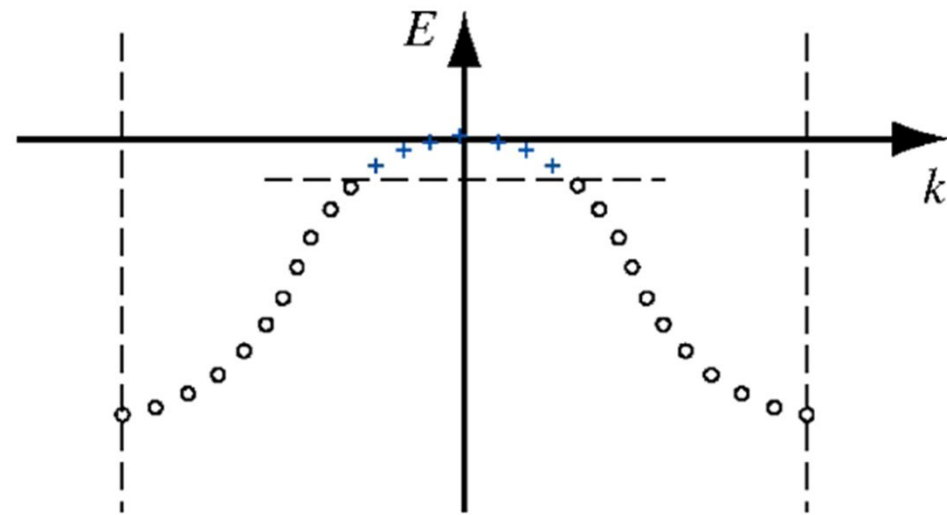


Hole: positive charges occupying the empty states



(a)

Valence band with conventional electron filled states and empty states



(b)

Concept of positive charges occupying the original empty states.

Effective mass of valence electrons

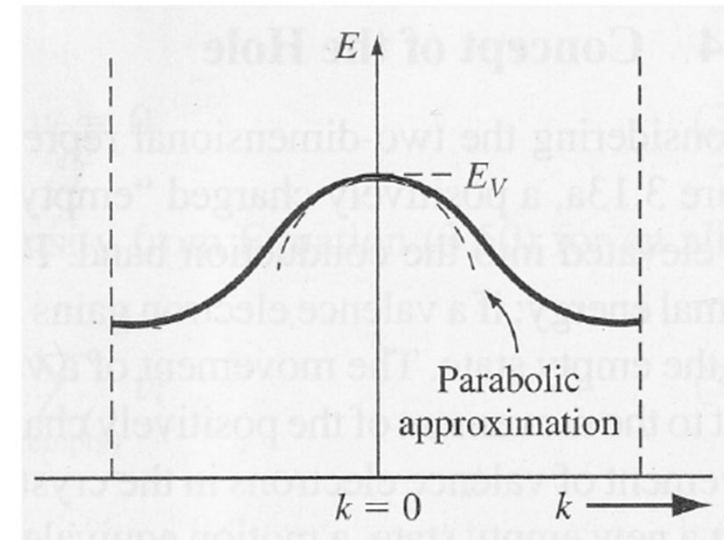
Energy at the top of allowed energy band can be written as

$$(E - E_v) = -C_2(k)^2$$

$$C_2 \rightarrow +ve$$

$$\frac{d^2 E}{dk^2} = -2C_2$$

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{-2C_2}{\hbar^2}$$



Compare with free electron:

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m}$$

Effective mass of valence electrons

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{-2C_2}{\hbar^2} = \frac{1}{m^*}$$

Negative effective mass

Bit strange!

$$F = m^* a = -eE$$

$$a = \frac{-eE}{-|m^*|} = \frac{+eE}{|m^*|}$$

- negative mass is a result of our attempt to relate quantum and classical mechanics. It is due to the inclusion of the effect of internal forces due to ions and other electrons.
- The net motion of electrons in the nearly full valence band can also be described by considering the empty states, provided +ve electronic charge and +ve effective mass is associated with them
- This new particle with +ve electronic charge and +ve effective mass denoted by m_p^* is called the hole

Comparison between electron and hole

Electrons:

- in almost empty band
- negative charge
- positive effective mass

Holes:

- in almost full band
- positive charge
- positive effective mass

Both:

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m^*}$$

$$F = m^* a = qE$$

Effective mass of electron and hole

	m_n^*/m_0	m_p^*/m_0
Silicon	1.08	0.56
Gallium Arsenide	0.067	0.48
Germanium	0.55	0.37

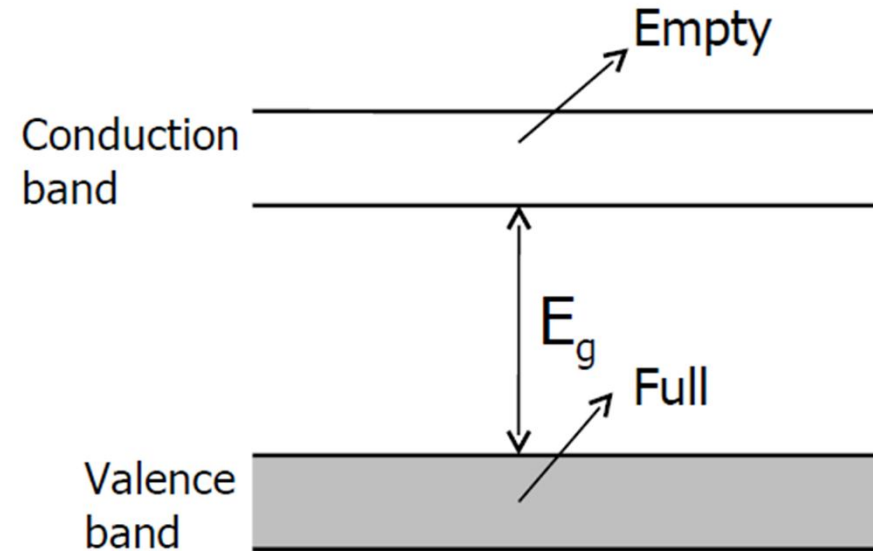
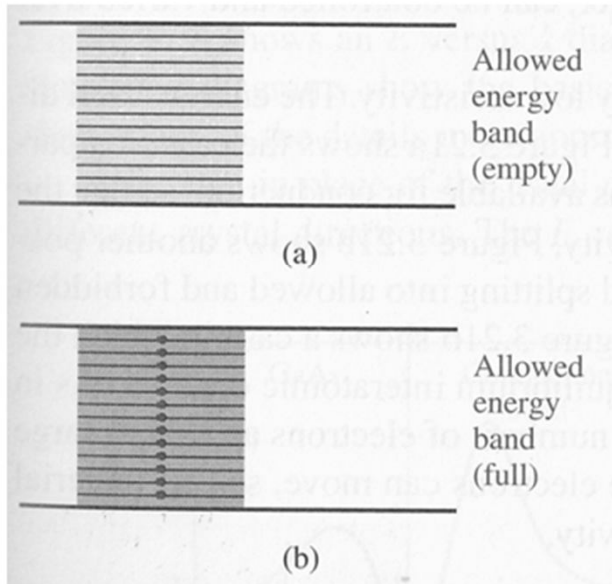
Table 4.2 Effective mass m_e^* of electrons in some metals

Metal	Ag	Au	Bi	Cu	K	Li	Na	Ni	Pt	Zn
$\frac{m_e^*}{m_e}$	0.99	1.10	0.047	1.01	1.12	1.28	1.2	28	13	0.85

Effective Mass	AlAs	GaAs	GaP	InP	InAs	InSb
m_e^* / m_0	0.124	0.067	0.5	0.079	0.024	0.014
m_{hh}^* / m_0	0.5	0.51	0.67	0.65	0.41	0.4

Energy band of insulators

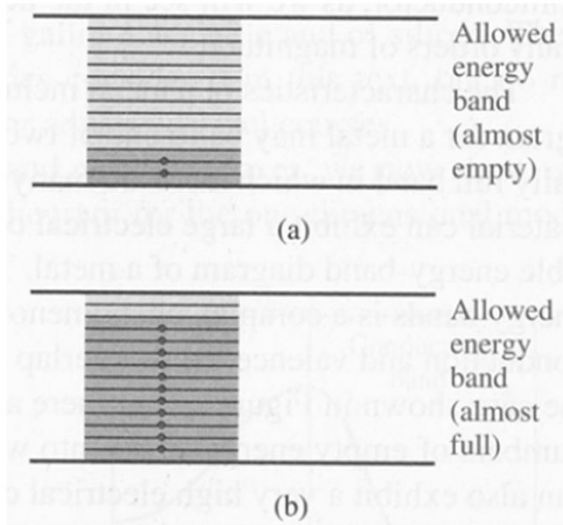
Insulators:



- E_g 3.5 to 6 eV or higher
- valence band full, conduction band empty
- no charged particles that contribute to drift current.

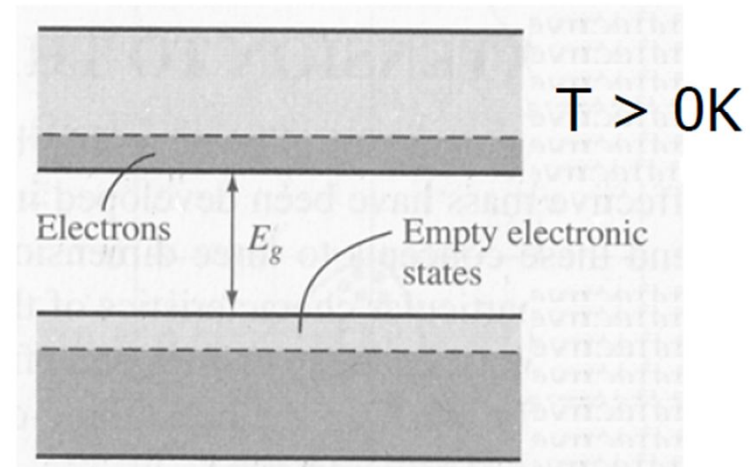
Energy band of semiconductors

Semiconductors:



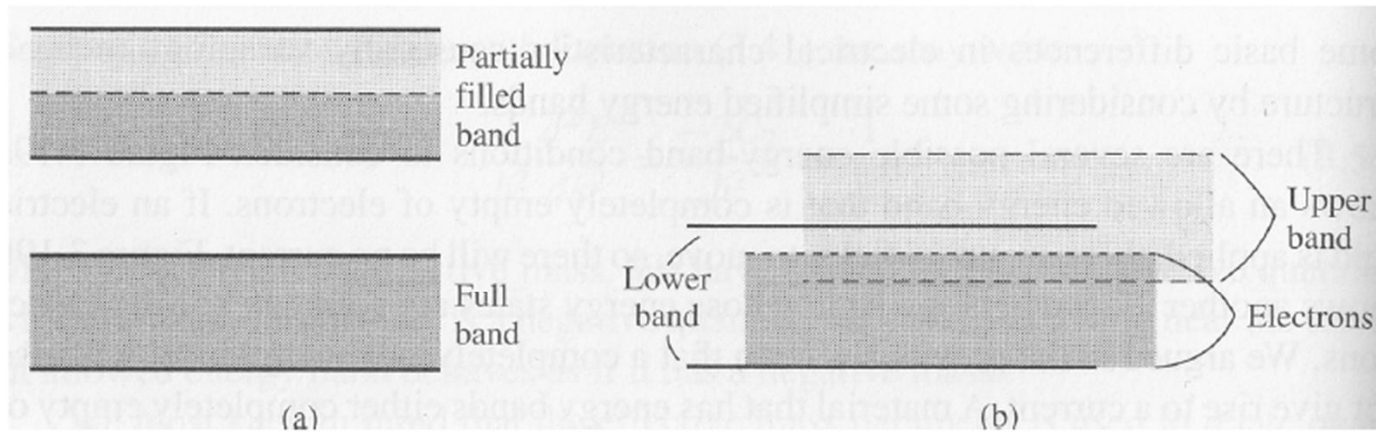
- E_g in semiconductors is small. $E_g \sim 1\text{eV}$
- At $T > 0\text{K}$, there will be some electrons in conduction band and holes in valence band, which can contribute to current.

- resistivity can be varied over many orders of magnitude by doping



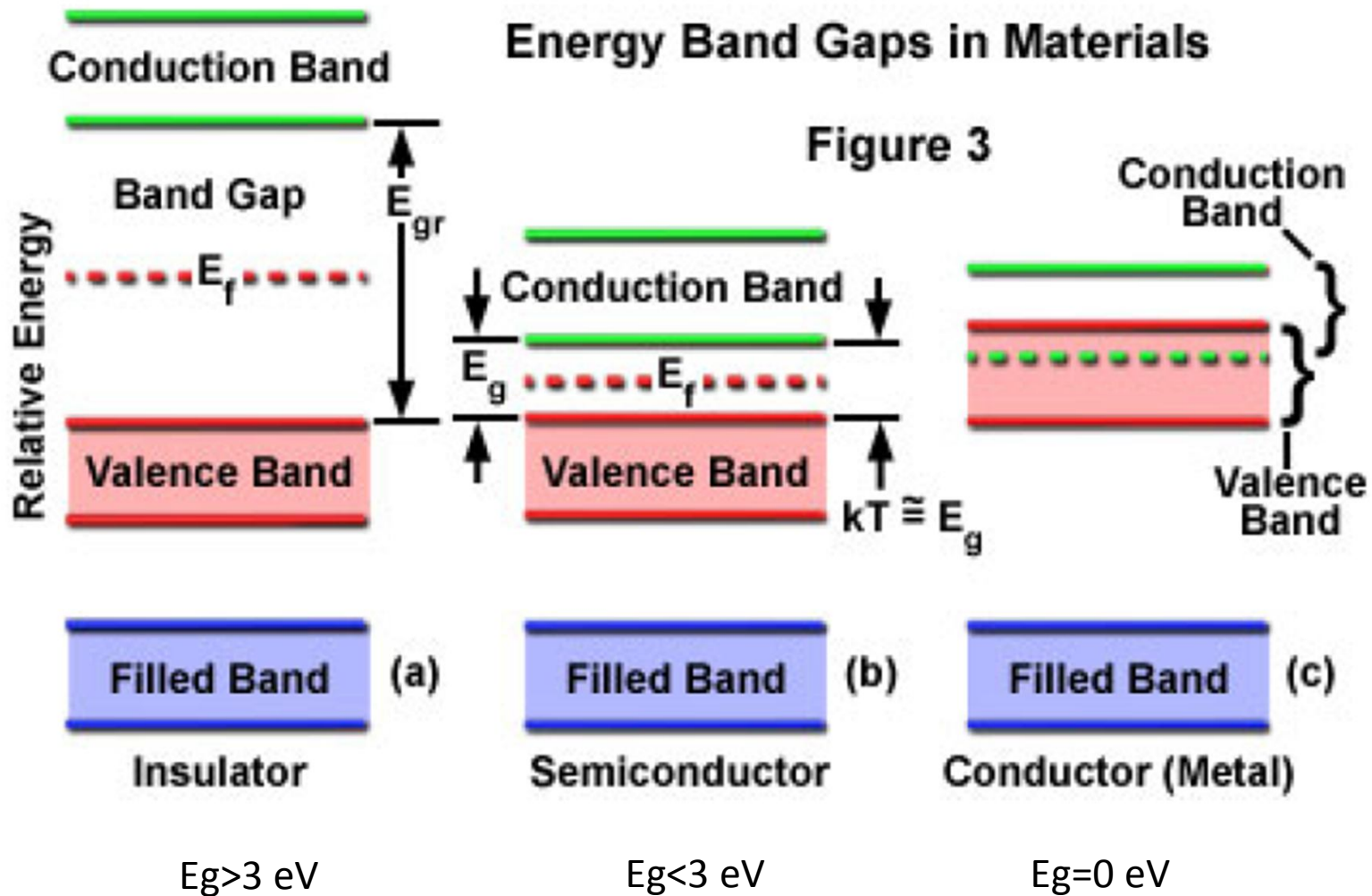
Energy band of metals

Metals:

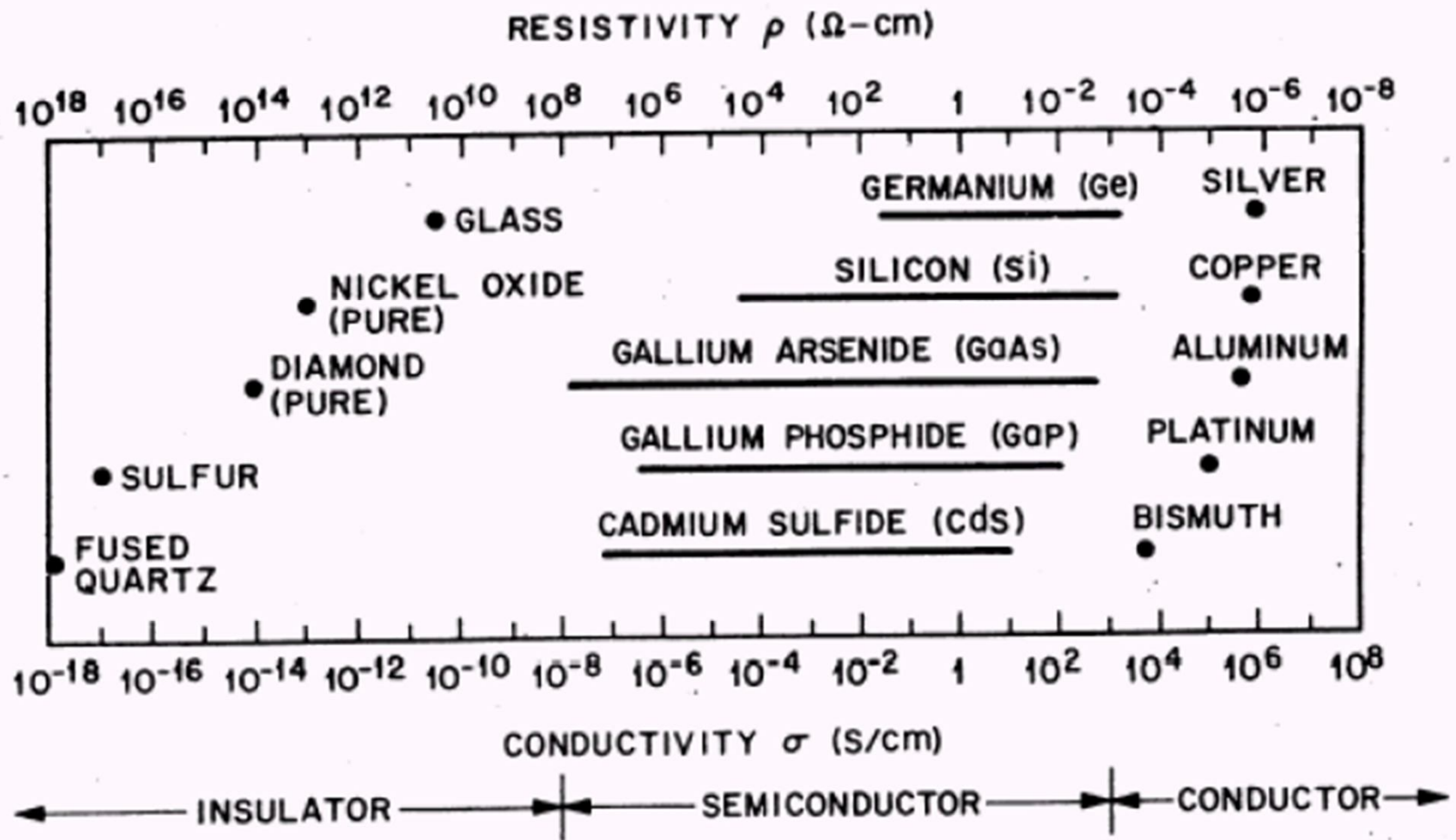


- Two possibilities - half filled conduction band or conduction and valence bands overlap.
- plenty of electrons and empty energy states that these electrons can move into.
- Very high electrical conductivity.

Insulator, semiconductor and metal



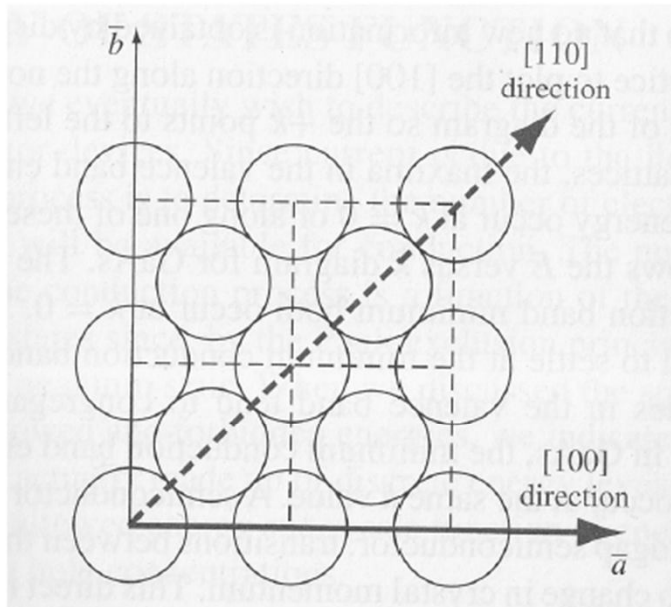
Insulator, semiconductor and metal



Typical range of conductivities for insulators, semiconductors, and conductors.

Extending energy band theory to three dimensions

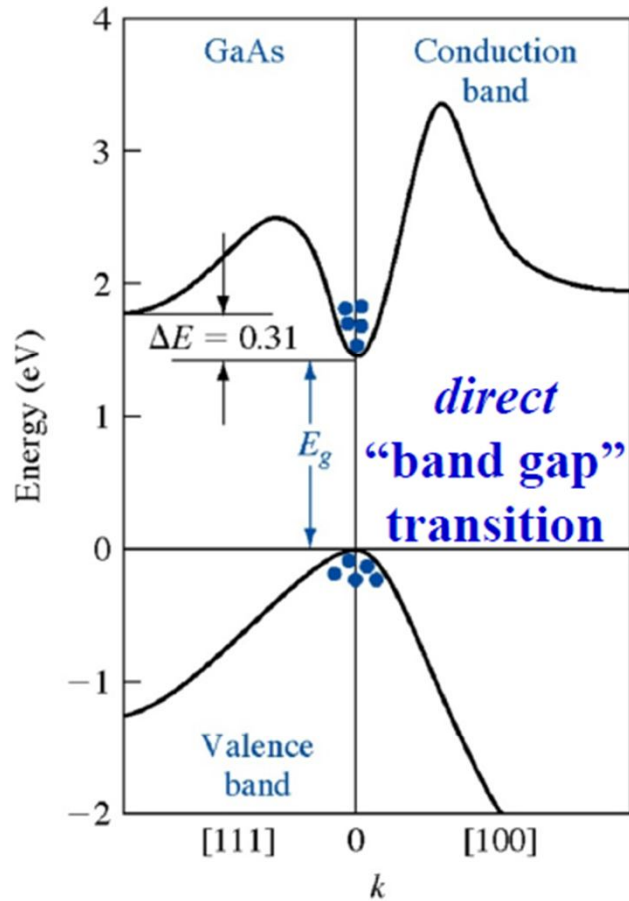
- Extend concepts of energy band and effective mass to 3-dimensions



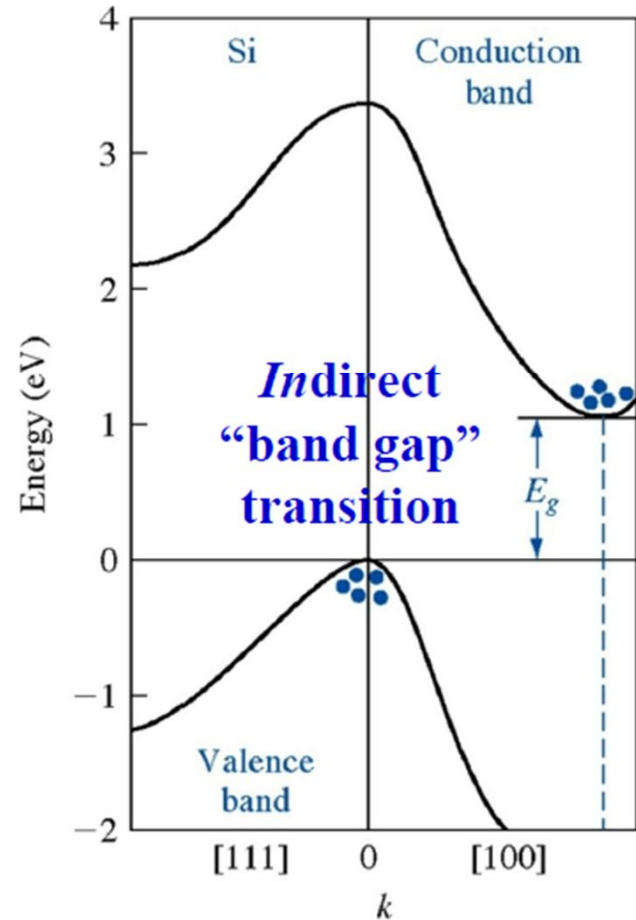
- distance between atoms not the same in different directions
- E vs. k diagram $\rightarrow$ as seen before depends on 'a' and 'a' varies with direction.

Different potentials in different directions

E vs. k diagrams of Si and GaAs



(a)



(b)

- [100] and [111] directions plotted along $+k$ and $-k$ axes.

E vs. k space diagrams of Si and GaAs

- In GaAs the minimum conduction band energy and maximum valence band energy occur at same k value. Such semiconductors are called *direct* band-gap semiconductors. Others are called *indirect* band-gap semiconductors. Eg. Si, Ge.
- Direct bandgap semiconductors are used for making optical devices. In indirect bandgap semiconductors there is loss of momentum during transition from conduction band to valence band, making them unsuitable for optical devices.

E vs. k space diagrams of Si and GaAs

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m^*} \longrightarrow \frac{d^2 E}{dk^2} \text{ is larger in GaAs than in silicon}$$

So m^* in GaAs < m^* in silicon

- effective mass can be different along 3 different k-vectors. We consider average effective mass hence forth.

Density of states function

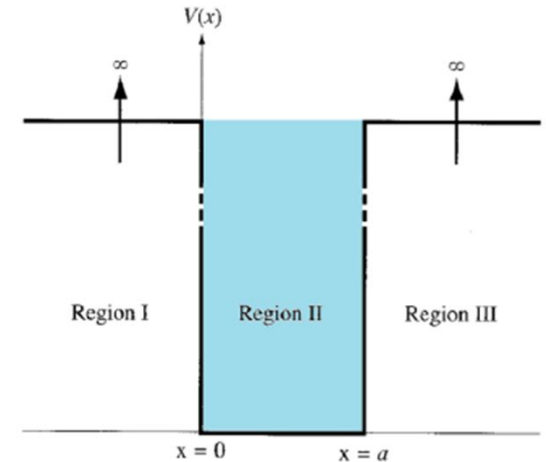
- we want to know the number of charge carriers and their temperature dependence
- hence the question is: how many energy levels do we have and what is the chance that they are populated in dependence of the temperature.

Mathematical model for density of states

- we consider a free electron confined to an 3-dimensional cubical infinite potential well of side 'a'.

$$V(x,y,z) = 0 \quad \text{for} \quad \begin{aligned} 0 < x < a \\ 0 < y < a \\ 0 < z < a \end{aligned}$$

$$V(x,y,z) = \infty \quad \text{elsewhere}$$



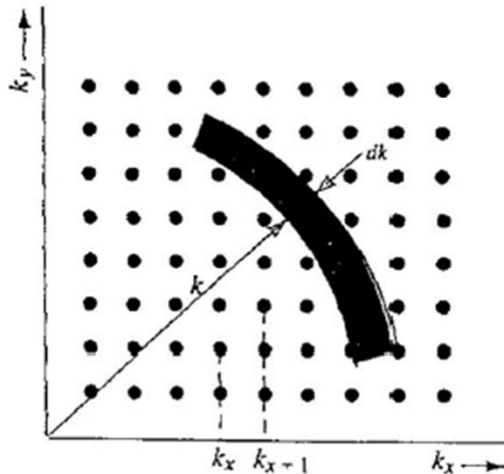
One dimensional potential well $\longrightarrow K = \frac{n\pi}{a}$ and $K = \sqrt{\frac{2mE}{\hbar^2}}$

Mathematical model for density of states

Extending to 3-dimensions,

$$\frac{2mE}{\hbar^2} = k^2 = k_x^2 + k_y^2 + k_z^2 = (n_x^2 + n_y^2 + n_z^2) \left(\frac{\pi^2}{a^2} \right)$$

Where n_x , n_y and n_z are positive integers (negative values lead to the same orbitals and hence do not represent a different quantum state)



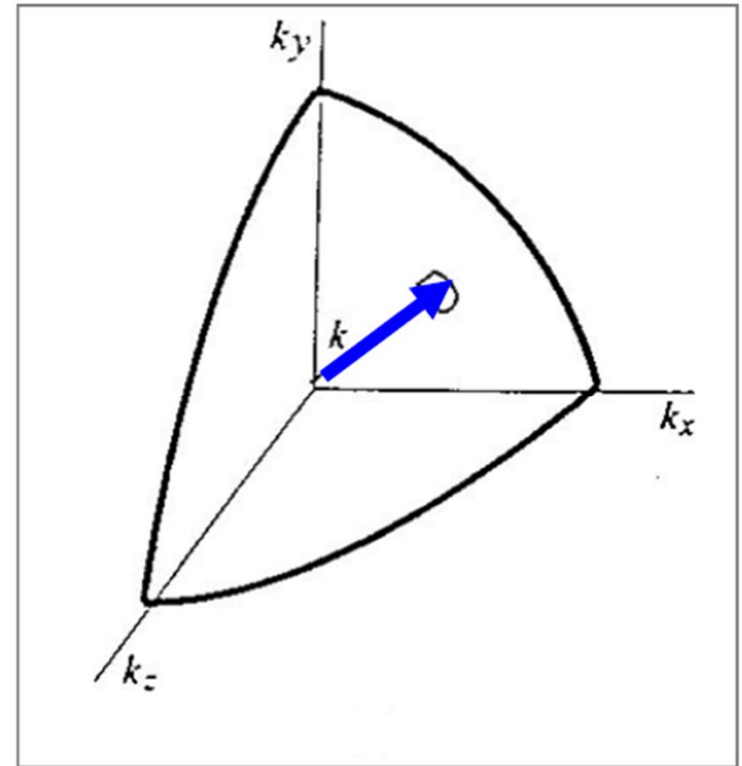
Two-dimensional array of allowed quantum states in k-space

Mathematical model for density of states

In 3-dimensional k-space, only $1/8^{\text{th}}$ of the spherical k-space needs to be considered for determining the density of states

-ve value of (k_x, k_y, k_z) result in same $k^2 = k_x^2 + k_y^2 + k_z^2$ and hence same energy. Since these do not result in separate energy state, they are not considered.

Spherical surface is considered because value of k^2 and hence energy is same along this surface.

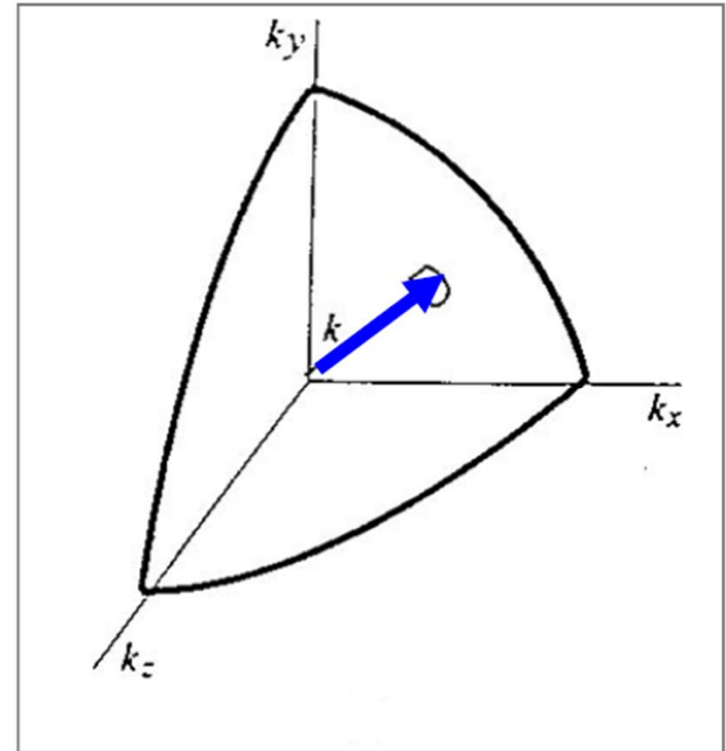


Positive $1/8^{\text{th}}$ of spherical k-space.

Mathematical model for density of states

Distance between quantum states in any one axial direction is -

$$k_{x+1} - k_x = (n_x + 1)\left(\frac{\pi}{a}\right) - n_x \left(\frac{\pi}{a}\right) = \frac{\pi}{a}$$



Positive 1/8th of spherical k-space.

Therefore Volume of a k -point: $V_k = \left(\frac{\pi}{a}\right)^3$

Mathematical model for density of states

Number of energy states
between $k, k+\Delta k$ $\longrightarrow$ $g_T(k)dk = 2 \left(\frac{1}{8}\right) \frac{4\pi k^2 dk}{\left(\frac{\pi}{a}\right)^3}$

Where:

$2 \rightarrow$ two spin states allowed for each quantum number

$4\pi k^2 dk \rightarrow$ differential volume in k -space between shells of radius k and $k+\Delta k$ $\frac{d}{dk}\left(\frac{4}{3}\pi k^3\right) = 4\pi k^2 dk$

Simplifying, $g_T(k)dk = \frac{\pi k^2 dk}{\pi^3} \cdot a^3$

Mathematical model for density of states

Use relationship between k & $E \rightarrow$ to move from k -space to real space.

$$k^2 = \frac{2mE}{\hbar^2} \longrightarrow k = \frac{1}{\hbar} \sqrt{2mE} \longrightarrow dk = \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE$$

Substituting for dk in previous equation,

$$g_T(E)dE = \frac{\pi a^3}{\pi^3} \left(\frac{2mE}{\hbar^2} \right) \cdot \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE$$

Mathematical model for density of states

$$g_T(E)dE = \frac{\pi a^3}{\pi^3} \left(\frac{2mE}{\hbar^2} \right) \cdot \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE$$


$$\square = \frac{h}{2\pi}$$

$$g_T(E)dE = \frac{4\pi a^3}{h^3} \cdot (2m)^{3/2} \cdot \sqrt{E} \cdot dE$$

$g_T(E)dE$ is the density of states in volume a^3 . Dividing by a^3 we can get the volume density of states as

$$g(E) = \frac{4\pi(2m)^{3/2}}{h^3} \sqrt{E}$$

Example

Example: calculate the density of states with energies between 0 and 1 eV for a free electron.

Solution:

$$\begin{aligned} N &= \int_0^E g(E) dE = \frac{4\pi(2m)^{3/2}}{h^3} \int_0^E \sqrt{E} dE \\ &= \frac{4\pi(2m)^{3/2}}{h^3} \frac{2}{3} E^{3/2} = \frac{8\pi(2mE)^{3/2}}{3h^3} \end{aligned}$$

$$\begin{aligned} N &= \frac{8\pi \left[2(9.11 \times 10^{-31})(1.6 \times 10^{-19}) \right]^{3/2}}{3(6.625 \times 10^{-34})^3} \\ &= 4.5 \times 10^{27} \text{ m}^{-3} = 4.5 \times 10^{21} \text{ cm}^{-3} \end{aligned}$$

Extension to semiconductors

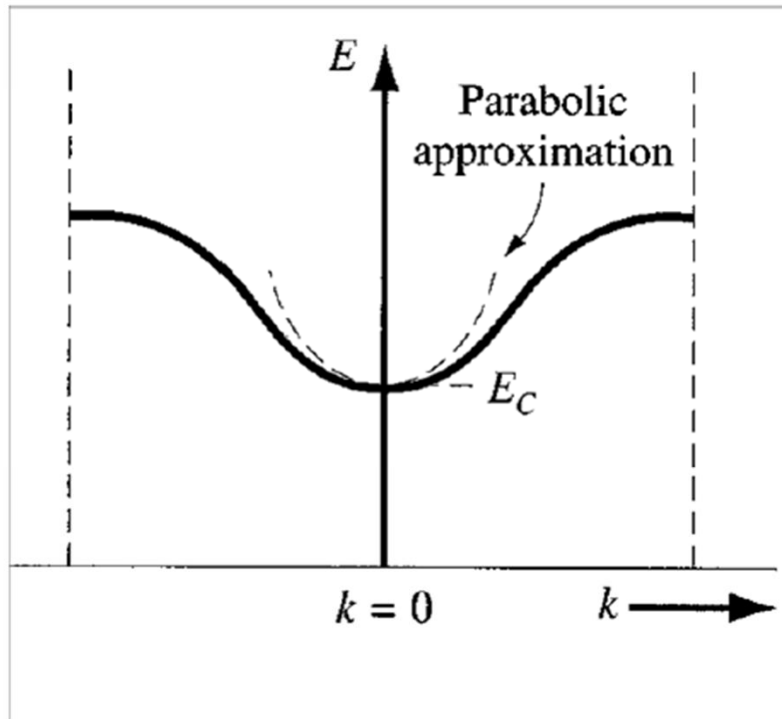
We have derived expression for density of allowed electron quantum states using model of free electron in an infinite potential well.

Electrons and holes are also confined within a semiconductor crystal

So we can extend this model to derive density of quantum states in conduction and valence bands.

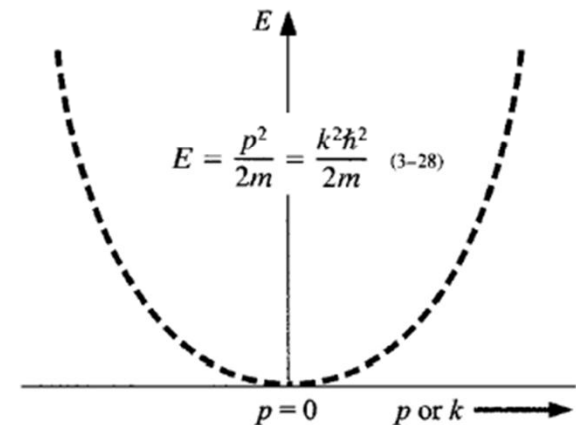
Extension to semiconductors

Conduction band:



$$E = E_c + \frac{\hbar^2 k^2}{2m_n^*}$$

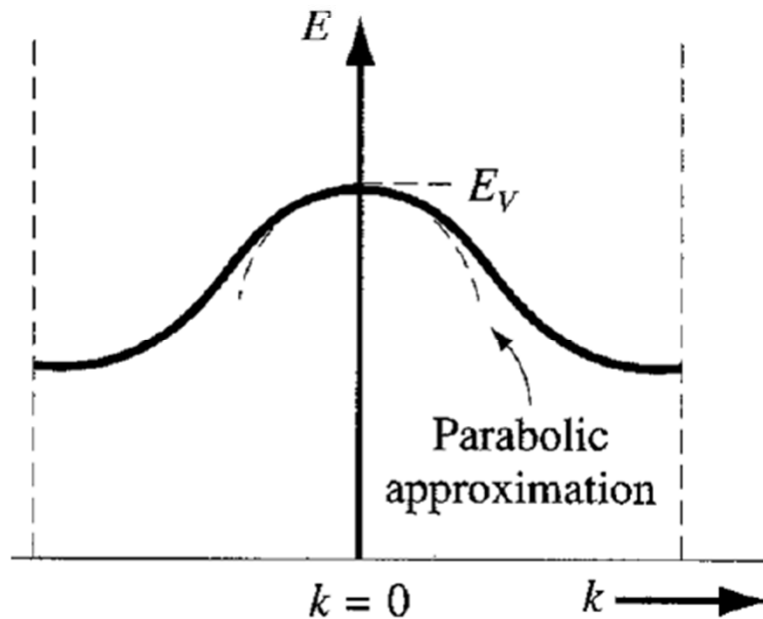
$$E - E_c = \frac{\hbar^2 k^2}{2m_n^*}$$



Parabolic E vs. k for free electron

Extension to semiconductors

Valence band:



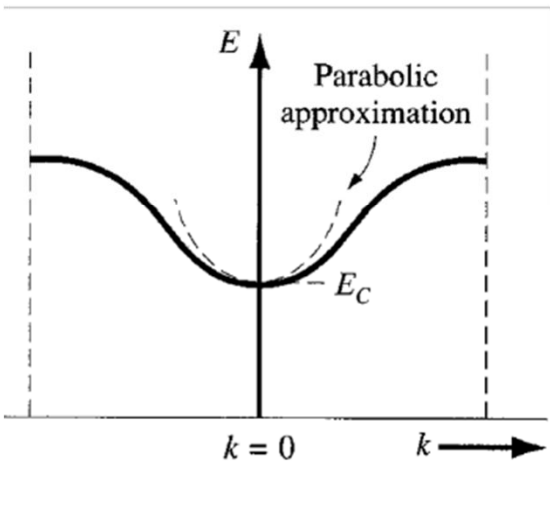
$$E = E_v - \frac{\hbar^2 k^2}{2m_p^*}$$

$$E_v - E = \frac{\hbar^2 k^2}{2m_p^*}$$

Density of states in the conduction band

For free electron, $E = \frac{\hbar^2 k^2}{2m}$ and $g(E) = \frac{4\pi(2m)^{3/2}}{h^3} \sqrt{E}$

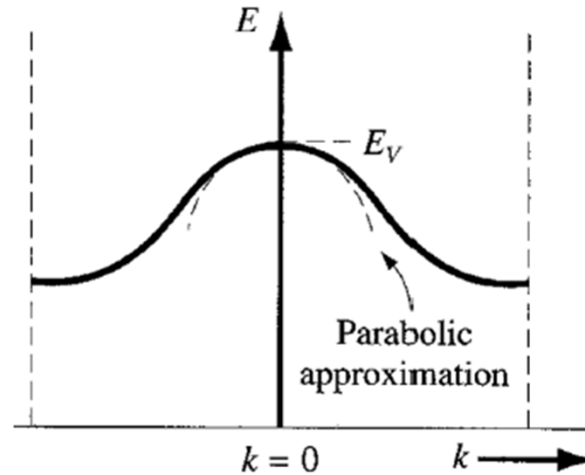
For conduction band, $E - E_c = \frac{\hbar^2 k^2}{2m_n^*}$



$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \sqrt{E - E_c}$$

Density of states in the valence band

for $E \leq E_v$



$$E_v - E = \frac{\hbar^2 k^2}{2m_p^*}$$

$$g_v(E) = \frac{4\pi (2m_p^*)^{3/2}}{h^3} \sqrt{E_v - E}$$

Density of states function

In general: $g(E) = \frac{4\pi(2m)^{3/2}}{h^3} \sqrt{E}$

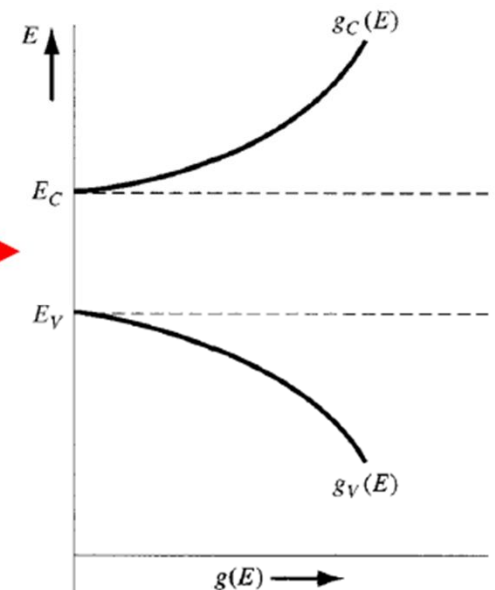
for $E \geq E_c$.

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \sqrt{E - E_c}$$

$$g(E) = 0 \text{ for } E_v < E < E_c \rightarrow$$

for $E \leq E_v$

$$g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3} \sqrt{E_v - E}$$



Statistical Mechanics

- While studying, large number of particles, we are interested only in statistical behavior of the group as a whole.
- In a semiconductor crystal, we are not interested in the behavior of each individual electron.
- The electrical characteristics will be determined by statistical behavior of large number of electrons.
- While studying statistical behavior, we must consider the laws that the particles obey.

Fermi-Dirac distribution function

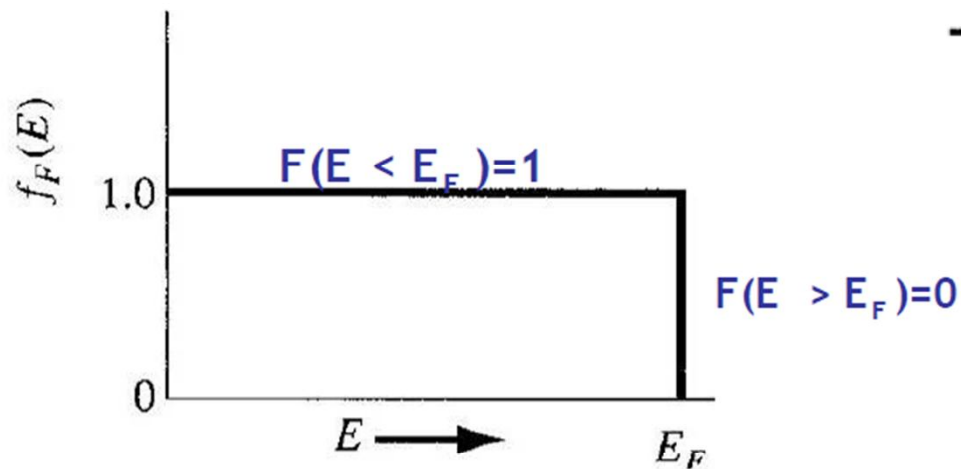
$$\frac{N(E)}{g(E)} = f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

$N(E) \rightarrow$ number of electrons per unit volume per unit energy.

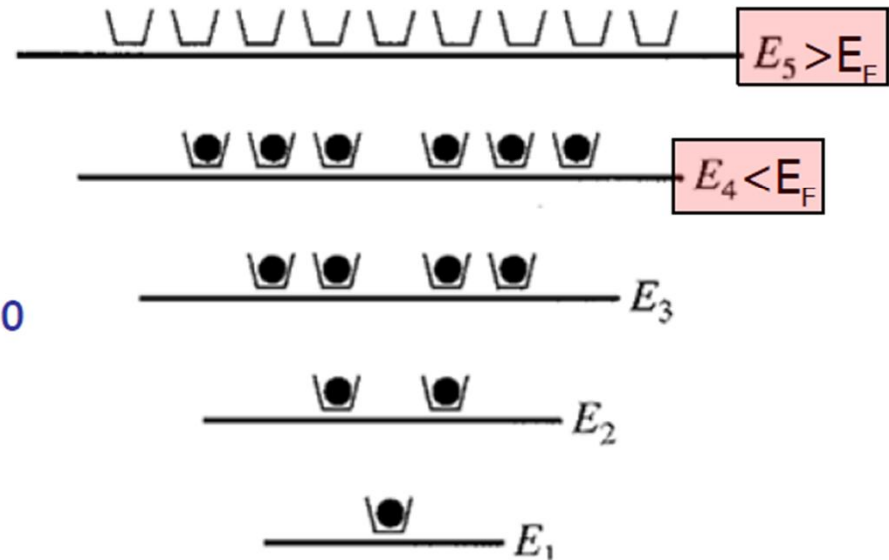
$g(E) \rightarrow$ number of quantum states per unit vol per unit energy.

$E_F \rightarrow$ Fermi energy (explained in more detail later)

At $T = 0$ K



The Fermi probability function versus energy for $T=0$ K



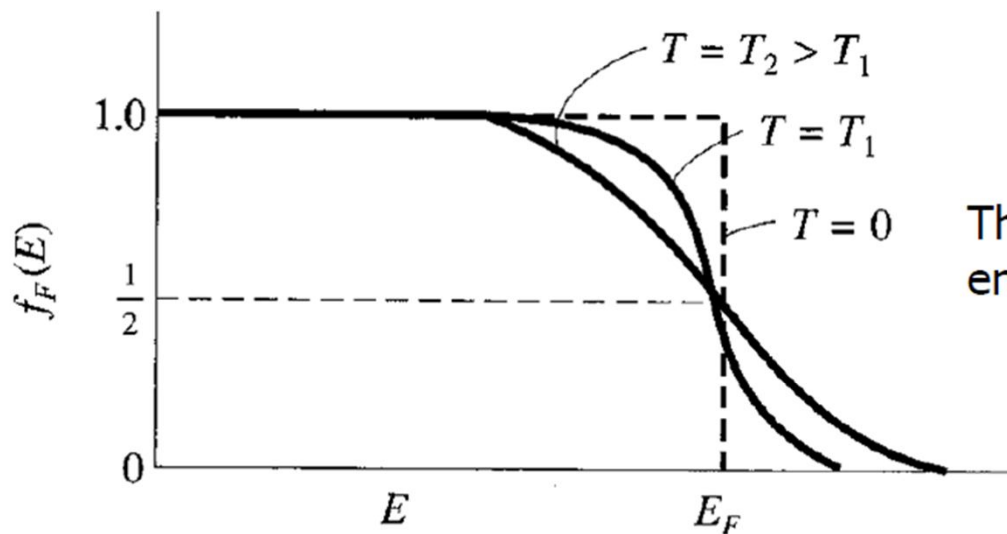
Discrete energy states and quantum states for a particular system.

At $E=E_F$

At $E=E_F$

$$f_F(E = E_F) = \frac{1}{1 + \exp(0)} = \frac{1}{1 + 1} = \frac{1}{2}$$

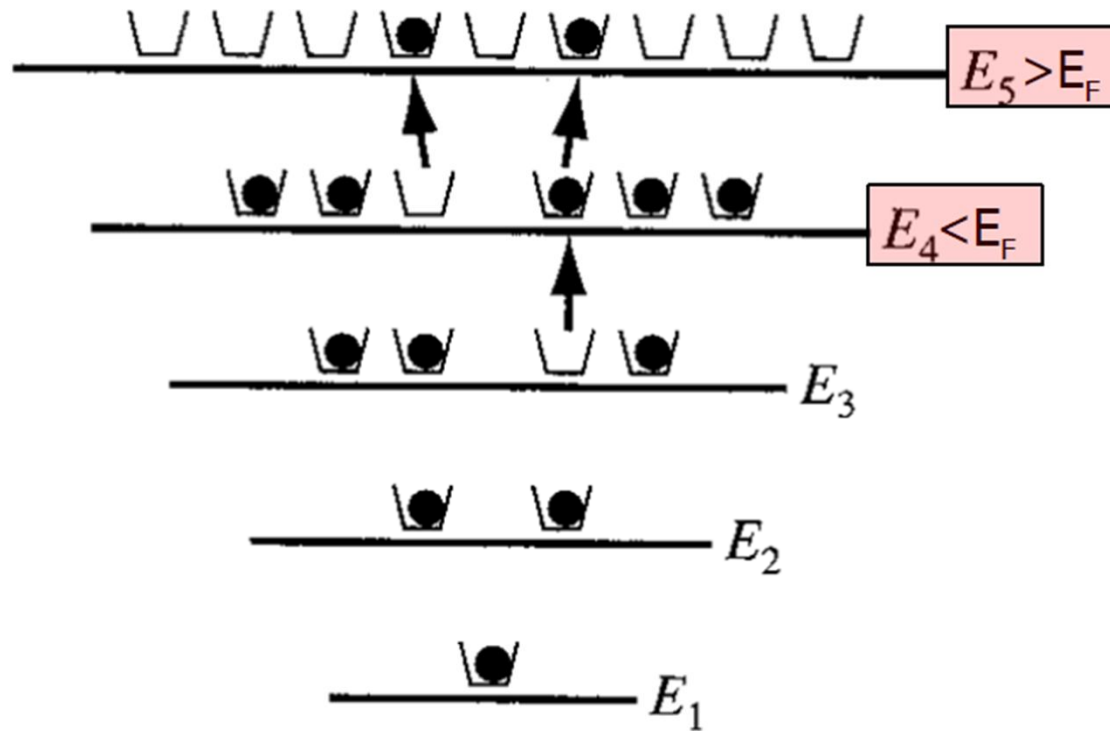
So Fermi Energy is the energy level where probability is $1/2$.
At $T > 0$



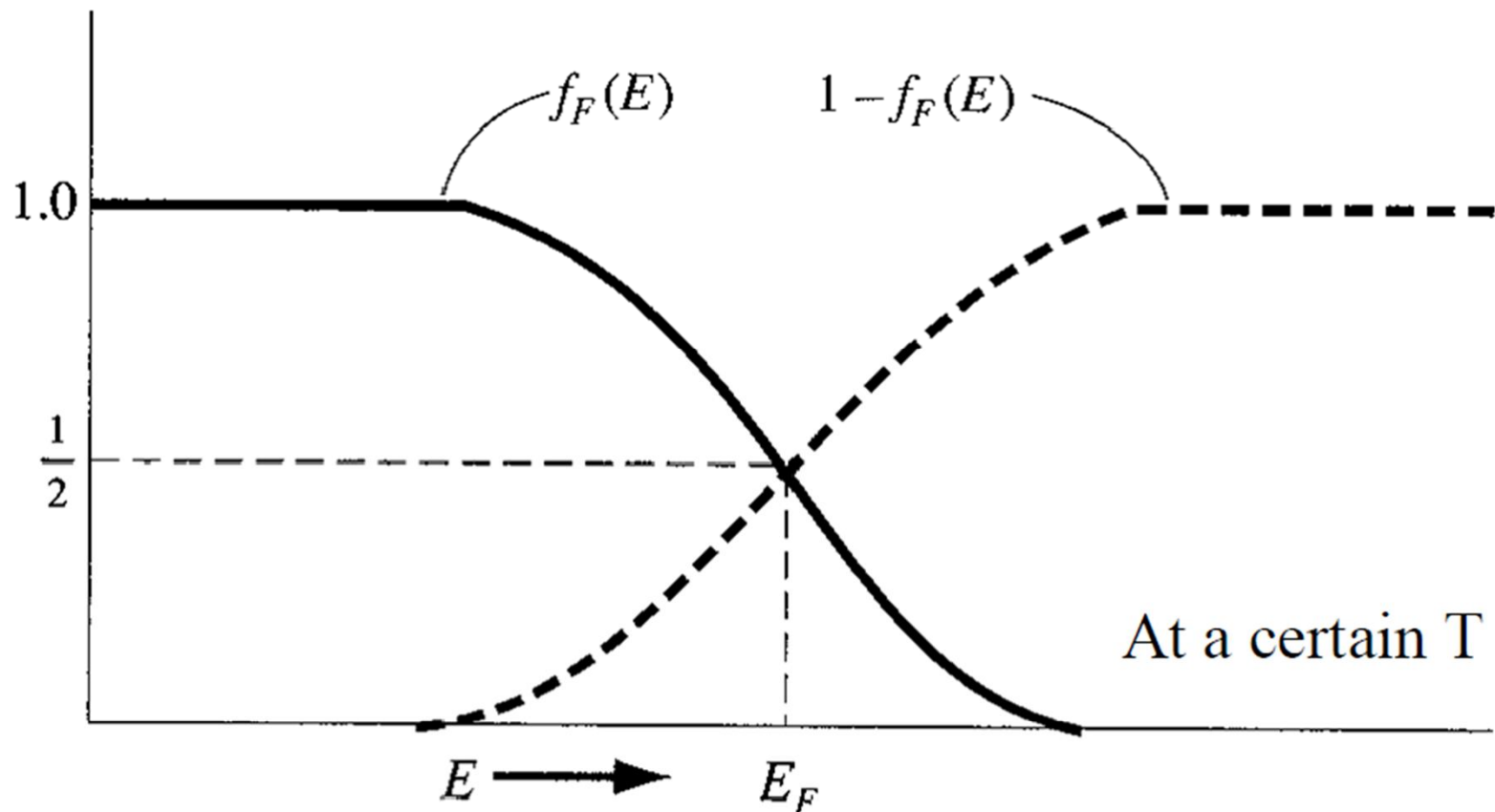
The fermi probability function versus energy for different temperatures.

At $T > 0K$, there is non-zero probability that some states above E_F are occupied and some states below are empty $\rightarrow$ some electrons have jumped to higher energy levels with increasing thermal energy.

At $T > 0$ K



Discrete energy states and quantum states for the same system at $T > 0$ K



The function $f_F(E)$ is symmetrical with the function $1 - f_F(E)$ about the Fermi energy E_F

Reading materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 3, “Introduction to the Quantum Theory of Solids”.

Q & A

- What are the different types of solids?
 - *Crystalline, amorphous, polycrystalline or*
 - *Semiconductor, metal, insulator.*
- What influences the arrangement?
 - *The outer shell electrons.*
 - *What states are available and occupied.*
- Why are the states in a solid different from those in isolated atoms or molecules?
 - *Pauli exclusion principle allows only one electron per state.*
- Why are two electrons allowed in an S-shell then?
 - *Electron spin up and spin down*

Q & A

- How do these states change when a large number of atoms are brought together?
 - They must split into multiple states so that only one electron will be in each state.
 - There are so many of these states that they appear to be a continuum, or a band of available energies.
- How does that arrangement influence the electrical characteristics of the material? What band configuration makes an insulator or semiconductor?
 - Insulators and semiconductors have a valence band and a conduction band separated by a band gap.

Q & A

- What is the difference in band structure between a semiconductor and an insulator?
 - Semiconductors have smaller band gaps ($\sim 1\text{eV}$) so are more likely to experience thermal or optical excitation into the conduction band.
- What do the energy bands of a metal look like?
 - Bands overlap to provide a continuum with no energy gap, i.e. no distinction between valence and conduction electrons, and
 - Conduction band is partially filled at non-zero temperatures.